Why did one alpha in eight thousand come back?

? A pudding atom should never turn a fast alpha around. What did the bounces actually prove?

1

Predict first, then look

Thomson spread the positive charge through a sphere of radius about 10^{-10} m. An alpha inside it feels a field growing linearly from the centre and peaking at kZe/R² at the surface. Work out the impulse before looking at the data: that is what makes this a test.

$$\theta \sim \frac{kZe^2}{KR}, \qquad \theta \lesssim 1^\circ$$

→ multiple scattering averages out further: the pudding predicts a straight beam

2

The impulse budget of a soft charge

A 5 MeV alpha spends about 10^{-18} s inside the atom, and the largest available force in that time gives a transverse momentum far below its own: a fraction of a degree. To turn it 90° the charge must sit inside 10^{-14} m, where the force is a hundred times larger.

$$f = \pi b^2 nt: \quad b_{90^\circ} \approx 2.3 \times 10^{-14}\ \text{m} \ \Rightarrow\ f \approx \frac{1}{8000}$$

→ f is the fraction of nuclei's cross-sectional area the foil presents

3

What the one-in-eight-thousand means

The observed fraction is an area measurement. Invert f = πb²nt and the angle-90° impact parameter falls out: nucleus tens of fm, atom 10^5 fm — a marble in an empty cathedral.

→ r_min is an upper bound on nuclear size while the force stays Coulomb — it is not a radius

4

A worked closest approach

A 5 MeV alpha on gold, head-on. All kinetic energy becomes potential energy at the turning point:

$$r_{\min} = \frac{2(79)(1.44\ \text{eV nm})}{5 \times 10^6\ \text{eV}} = 4.55 \times 10^{-5}\ \text{nm}$$

$$r_{\min} = 45.5\ \text{fm} \quad \text{against } R \approx 7\ \text{fm}$$

→ six nuclear radii of clearance: the alpha never touched the nucleus, so the Coulomb law held

KEY RESULT

the two numbers

$$r_{\min} = \frac{2kZe^2}{K}$$

$$\tan\frac{\theta}{2} = \frac{kZe^2}{mv^2 b}$$

f = πb²nt is an area ratio, not a force

TRAP

what r_min is not

It is not the nuclear radius. It is a head-on upper bound, valid only while the force is pure Coulomb.

Large-angle scattering does not mean contact: it means small b.

NUMBERS

carry these

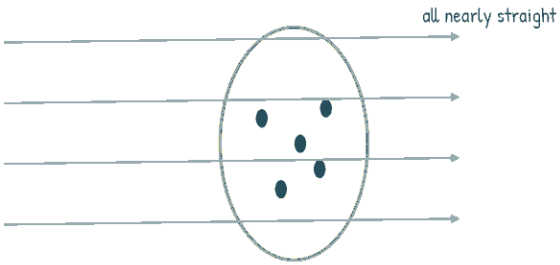
1 in 8000 back-scattered (gold, ~90°)

kZe² for gold = 114 eV nm

nuclear scale few fm; atom 10^5 fm

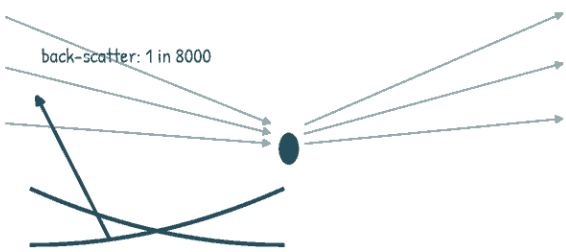
Same beam, two predictions: the rare back-scatter is the test that decides

a soft cloud cannot turn a fast alpha



Thomson: charge spread over 10^-10 m

a point charge can turn it around



Rutherford: one point charge, sin^-4 law

KEEP / the rare back-scatter is the measurement: charge sits in a nucleus, not a cloud

How close can an alpha get before the law breaks?

? Rutherford's formula held at every angle measured. Where must it stop, and what stops it?

1 The hyperbola, in one line

A Coulomb encounter is a conic section. The impact parameter b is the miss distance the alpha would have had with no force at all; the scattering angle follows from angular momentum plus the impulse integral.

$$\tan\frac{\theta}{2}=\frac{kZe^2}{mv^2 b}=\frac{r_h}{2b},\qquad r_h=\frac{2kZe^2}{K}$$

→ b → 0 gives back-scatter; b → ∞ gives a straight track

2 Invert it and you have a ruler

Because the angle depends on b only through the ratio r_h/2b, measuring θ measures b. The fraction scattered beyond a given angle is then the area of the disk πb² times the areal density of nuclei, nt.

$$\frac{d\sigma}{d\Omega}=\left(\frac{r_h}{4}\right)^2\frac{1}{\sin^4(\theta/2)}$$

→ the $\sin^{-4}$ law: halving the angle multiplies the count sixteenfold

3 Where the Coulomb picture must fail

Set r_min equal to the nuclear radius. Above that beam energy the alpha touches the nuclear surface, the potential stops being 1/r, and the measured distribution departs from $\sin^{-4}$. Natural alphas at 5–8 MeV probe only the clean Coulomb regime; accelerators above a few tens of MeV see the deviation.

$$K_{\text{fail}}=\frac{2kZe^2}{R}\approx 32\ \text{MeV}\quad(\text{gold},\ R\approx 7\ \text{fm})$$

4 A worked 30 MeV probe

Raise the beam to 30 MeV on the same gold foil and see how much room is left:

$$r_{\min}=\frac{227.5\ \text{eV nm}}{3\times10^7\ \text{eV}}=7.6\times10^{-6}\ \text{nm}$$

$$=7.6\ \text{fm}\approx R_{\text{Au}}$$

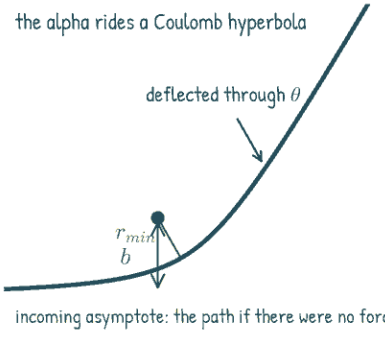
→ at about 32 MeV the alpha touches the surface and $\sin^{-4}$ scattering starts to fail

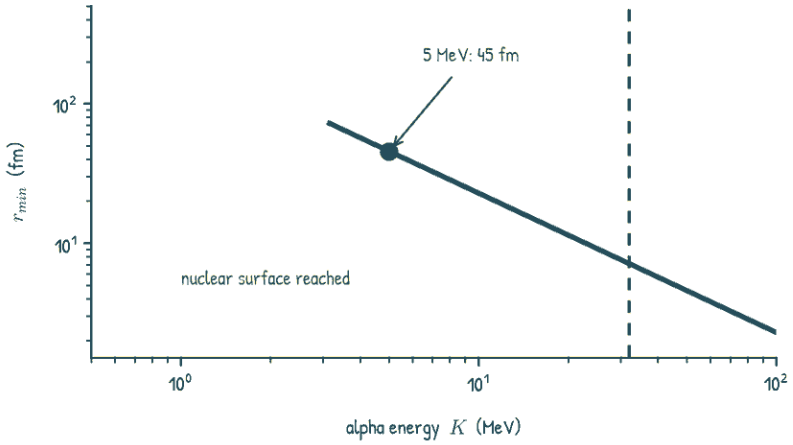
KEY RESULT
the geometry
$\tan(\theta/2)=r_h/2b$
$r_h=2kZe^2/K$
At 90°: b = r_h/2, always.

GATE
when the formula holds
Pure Coulomb, point nucleus, no recoil, non-relativistic alpha.
All four fail at high energy or high Z.

TRAP
angle convention
θ is measured from the incoming direction, not from the normal to the foil.

r_min is a head-on upper bound on the nuclear size, not a radius measurement





KEEP / r_min shrinks as 1/K until it reaches the nuclear surface — that is where new physics starts

▼13.6 eV
0.1 eV 10 MeV

Why is angular momentum n times h-bar?

? Bohr postulated it. de Broglie's wave makes it a closure condition on a circle.

1

A wave on a ring either closes or cancels

Put a de Broglie wave on a circle. If a whole number of wavelengths fits the circumference it meets itself in phase and stands; if not, it destroys itself. That single condition is the quantisation.

$$2\pi r = n\lambda = n\frac{h}{p} \;\Rightarrow\; pr = n\hbar$$

→ on a circle, pr is exactly the angular momentum — nothing else had to be assumed

2

Add the force, get the atom

Two equations now: closure and Coulomb force balance. Eliminate v between them and every hydrogen number follows with no adjustable constant.

$$\frac{mv^2}{r} = \frac{ke^2}{r^2}, \qquad mvr = n\hbar$$

$$r_n = \frac{n^2\hbar^2}{m_e k e^2} = 0.529\,\frac{n^2}{Z}\ \text{Å}$$

3

The same algebra gives speed and energy

Substituting back gives the speed and, with E = K + U, the ladder. K = −E and U = 2E, so the virial relation is a free check on every result.

$$v_n = \frac{kZe^2}{n\hbar} = 2.19\times10^6\,\frac{Z}{n}\ \text{m/s}$$

$$E_n = -\frac{m_e k^2 Z^2 e^4}{2\hbar^2 n^2} = -13.6\,\frac{Z^2}{n^2}\ \text{eV}$$

→ K = −E and U = 2E, so the electron is always more kinetic than bound

4

A worked wavelength check

The n = 3 orbit of hydrogen, checked against de Broglie directly:

$$v_3 = 7.29\times10^5\ \text{m/s}, \qquad \lambda = \frac{h}{m_e v_3} = 9.97\ \text{Å}$$

$$3\lambda = 29.9\ \text{Å} = 2\pi(4.76\ \text{Å})\ \checkmark$$

→ a 10 % error in n would leave the wave 36 degrees out of phase, cancelling itself

KEY RESULT

the closure

$$2\pi r = n\lambda$$

$$L = n\hbar$$

One condition replaces a postulate.

INSIGHT

why it is exact

The condition fixes the shape of the wave, and the Coulomb force fixes its scale. Two equations, two unknowns, no fitting.

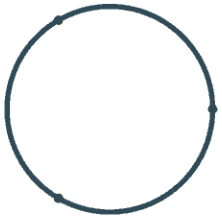
TRAP

the two-pi slip

L = nh/(2π) = nℏ, never nh. And λ = h/p, not h/mv with a different v.

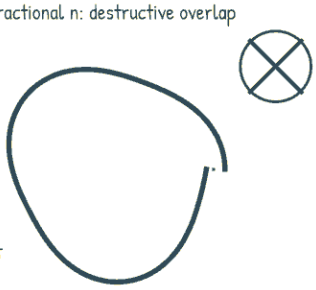
Bohr quantisation is a standing-wave condition: L = n h-bar is 2 pi r = n lambda

3 whole wavelengths close in phase



$2\pi r = n\lambda$

fractional n: destructive overlap



no closed wave, no orbit

KEEP / an orbit is allowed when the electron's wave closes on itself in phase

▼13.6 eV
0.1 eV 10 MeV

What does the hydrogen ladder actually look like?

? Thirty years of spectroscopy in one diagram. Read it as a table, not a picture.

① Three columns, one formula each

Every hydrogenic quantity is a power of n. The radius grows as n², the speed falls as 1/n, and the binding energy falls as 1/n². Write them as three columns and the page becomes a lookup table rather than a derivation.

$$r_n = 0.529\frac{n^2}{Z}\ \text{Å}, \quad v_n = 2.19\times10^6\frac{Z}{n}, \quad E_n = -13.6\frac{Z^2}{n^2}\ \text{eV}$$

② The gap, not the level, is what you see

A photon carries the difference between two levels. Using hc = 1240 eV nm, a gap in eV converts to a wavelength by one division, which is the whole arithmetic of the chapter.

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right), \qquad \lambda = \frac{1240}{\Delta E}$$

→ R = 1.097×10⁷ m⁻¹; hcR = 13.6 eV, so the two forms are the same statement

③ Bound means negative

Zero energy is the electron at rest at infinity. A bound electron therefore has negative total energy, and its ionisation energy is the positive quantity −E_n. Excitation energy is the difference of two levels; ionisation always ends at zero.

$$\text{IE}_n = -E_n = \frac{13.6\,Z^2}{n^2}\ \text{eV}, \qquad \text{IP (volts)} = \text{IE (eV)}$$

→ for a singly charged electron the two numbers coincide; for an alpha they do not

④ A worked ionisation ladder

Ionise hydrogen from n = 2, and then from n = 3:

$$\text{IE}_2 = \frac{13.6}{4} = 3.40\ \text{eV}, \qquad \text{IE}_3 = \frac{13.6}{9} = 1.51\ \text{eV}$$

$$\lambda_{\text{edge}} = \frac{1240}{3.40} = 365\ \text{nm}$$

→ the Balmer edge at 365 nm is the same number seen as a wavelength

KEY RESULT
the ladder

$$E_n = -13.6\,Z^2/n^2$$

$$\lambda = 1240/\Delta E\ (\text{nm, eV})$$

r ∝ n², v ∝ 1/n, |E| ∝ 1/n²

CHECK
instant audits

Ground state must return −13.6 eV at Z = 1.

A gap must be smaller than the ionisation energy it sits under.

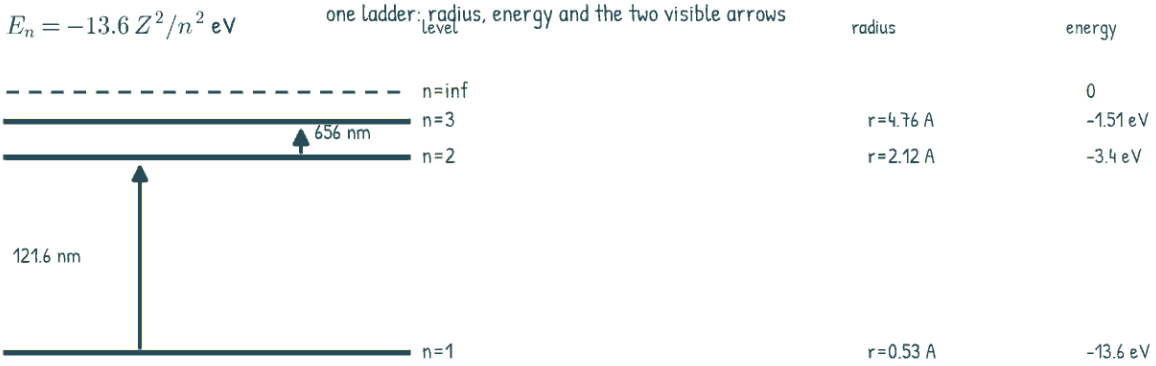
TRAP
signs and units

Subtracting two excited levels when asked to ionise — ionisation ends at 0 eV.

Plugging a wavelength in nm into hc = 1240 eV nm the wrong way round.

One formula per column: radius grows as n squared, binding falls as 1/n squared

$E_n = -13.6\,Z^2/n^2$ eV



KEEP / radius as n², binding as 1/n², and 13.6(1/n_f² − 1/n_i²) eV per arrow

▼1.89 eV
0.1 eV 10 MeV

Which series is which, and why five names?

? Lyman, Balmer, Paschen, Brackett, Pfund: one formula, five finishing rungs.

① **Fix the lower level and the family appears**

Everything that lands on n = 1 is Lyman (ultraviolet), on n = 2 is Balmer (the visible one), on n = 3 is Paschen (infrared), then Brackett and Pfund further out. The formula does not change; only n_f does.

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right), \qquad n_i > n_f$$

② **The two ends of any series**

The longest wavelength in a series is the smallest step, n_f + 1 → n_f. The shortest is the limit n_i → ∞, which is the ionisation edge of that lower level. Every series therefore runs from a line to a continuum edge.

$$\lambda_{\max} = \frac{1}{R}\frac{n_f^2(n_f+1)^2}{2n_f+1}, \qquad \lambda_{\min} = \frac{n_f^2}{R}$$

→ Balmer: 656 nm down to the 365 nm edge, which is why it is the visible one

③ **What the eye can and cannot see**

Only Balmer overlaps the 380–750 nm window, and only its first three lines are comfortably visible. Lyman is vacuum ultraviolet; Paschen is infrared. This is not an accident of hydrogen — it is why stellar spectra are classified by which series is strong.

→ a hydrogen discharge looks pink because Hα at 656 nm dominates the visible output

④ **A worked series limit**

The two ends of the Balmer series, computed from the same rule:

$$\lambda_{\max} = \frac{1}{R}\frac{4\cdot 9}{5} = 656 \text{ nm } (n_i = 3)$$

$$\lambda_{\min} = \frac{4}{R} = 364.6 \text{ nm } (n_i \to \infty)$$

→ everything between is visible; beyond 364.6 nm the series ends in a continuum

KEY RESULT
the five names
Lyman 1, Balmer 2, Paschen 3, Brackett 4, Pfund 5.
Only Balmer is visible.

$$\lambda_{\min} = n_f^2/R$$

TRAP
counting the wrong thing
"Lines in the Balmer series from n = 5" is four (5→2, 4→2, 3→2 plus the 5→… no: 5→2, 4→2, 3→2 = three). Total lines from n = 5 is ten.

NUMBERS
bench numbers
Hα 656.3 nm · Hβ 486.1 · Hγ 434.0 · Hδ 410.2
Lyman α 121.6 nm · Balmer edge 364.6 nm

Every series is the same Rydberg rule with a different finishing rung



one ladder, three series, one visible window



KEEP / the series is named by its lower level: n_f = 1, 2, 3, 4, 5

What can a 12.5 eV electron do that a 12.5 eV photon cannot?

? Same energy, different menu. The rule about who may pay in part is worth marks every year.

1 ## The electron menu

A bombarding electron can transfer any amount up to its whole kinetic energy, because it keeps the remainder as its own motion. So every level with excitation energy below the beam energy is reachable.

$$E_{\text{exc}} = 13.6\left(1 - \tfrac{1}{n^2}\right)\text{eV}: \;\; 10.2,\;\; 12.09,\;\; 12.75$$

→ a 12.5 eV beam reaches n = 2 and n = 3, and stops short of n = 4

2 ## The photon menu

A photon is created or absorbed as one unit. It cannot give 10.2 eV and carry 2.3 eV away, so a 12.5 eV photon is not absorbed by the 10.2 eV transition at all — it simply passes. Resonance is a matching condition, not a threshold.

$$hf = E_i - E_f \quad \text{exactly}$$

→ this is why the absorption spectrum has dark lines at precise wavelengths only

3 ## Above the ionisation limit

Once the photon energy exceeds the binding energy the electron is freed and the surplus becomes kinetic energy. The spectrum changes character: discrete lines below the limit, a continuum above it. The ionisation edge is the last line of a series.

$$K_{\max} = hf - \text{IE}, \qquad hf > \text{IE}$$

4 ## A worked threshold ladder

Which levels can a 12.5 eV electron reach in hydrogen?

$$E_2 = 10.2, \quad E_3 = 12.09, \quad E_4 = 12.75 \text{ eV}$$

$$12.5 > 12.09 \text{ but } 12.5 < 12.75 \;\Rightarrow\; n \le 3$$

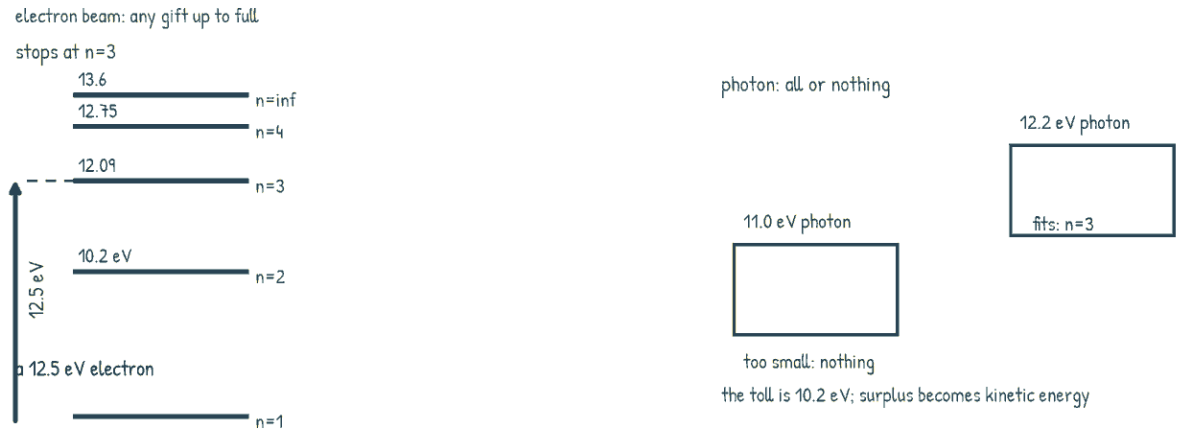
→ and the surplus 0.41 eV stays with the electron as kinetic energy

KEY RESULT
the asymmetry
$$E_{\text{exc}} = 13.6(1 - n^{-2})$$
Electron: any gift up to full. Photon: all or nothing.

TRAP
three classics
Letting an 11 eV photon excite the 10.2 eV level.
Forgetting the cascade: exciting n = 3 gives three lines, not one.
Confusing excitation energy with ionisation energy.

NUMBERS
hydrogen thresholds
n=2: 10.2 eV / 121.6 nm
n=3: 12.09 eV / 102.6 nm
ionisation: 13.6 eV / 91.2 nm

Electrons may pay in part; photons must pay in full - that is the excitation asymmetry



KEEP / electrons may hand over part of their energy; photons must be absorbed whole

How far does Z² scaling carry you?

? He^+, Li^{2+}, muonic hydrogen and positronium are the same atom with two numbers changed.

1

Charge: divide the radius, square the energy

A nucleus of charge Ze pulls harder, so the orbit shrinks as 1/Z while every energy multiplies by Z². He^+ therefore has a ground state at −54.4 eV and an orbit of 0.265 Å; Li^{2+} has −122.4 eV.

$$r_n \propto \frac{1}{Z}, \qquad E_n \propto Z^2, \qquad \lambda \propto \frac{1}{Z^2}$$

→ the 2 → 1 line of Li^{2+} is 13.5 nm, deep in the soft X-ray

2

Mass: the nucleus is not nailed down

The electron and nucleus orbit their common centre of mass, and every mass-dependent result takes the reduced mass μ = m_eM/(m_e+M). For hydrogen the correction is 0.05 %; for deuterium it shifts H^α by 0.18 nm, which is how deuterium was found.

$$R_M = R_\infty \frac{\mu}{m_e}, \qquad \Delta\lambda/\lambda \approx \frac{m_e}{2M}$$

3

Exotic atoms are scaling tests

Replace the electron with a muon and μ ≈ 207 m_e: the atom shrinks by 207 and binds 207 times harder, so its orbit sits inside the nucleus and its levels measure nuclear size. Positronium has μ = m_e/2, so it is twice as big and half as bound.

$$a_\mu = \frac{0.529\ \text{Å}}{207} = 256\ \text{fm}, \qquad E_1 = -2.81\ \text{keV}$$

→ positronium: 1.06 Å, −6.8 eV, and every wavelength doubled

4

A worked Li2+ line

The 2 → 1 resonance line of Li^{2+} (Z = 3):

$$\Delta E = 13.6(9)\left(\tfrac{1}{1} - \tfrac{1}{4}\right) = 91.8\ \text{eV}$$

$$\lambda = \frac{1240}{91.8} = 13.5\ \text{nm}$$

→ soft X-ray, and exactly one ninth of hydrogen's 121.6 nm: λ ∝ 1/Z²

KEY RESULT

the two dials

$$r \propto n^2/Z$$

$$E \propto Z^2/n^2$$

and m_e → μ for any mass-dependent quantity

TRAP

forgetting Z

Using −13.6/n² for He^+. Every energy carries Z², every radius 1/Z.

Using m_e where the reduced mass belongs — the whole point of exotic atoms.

NUMBERS

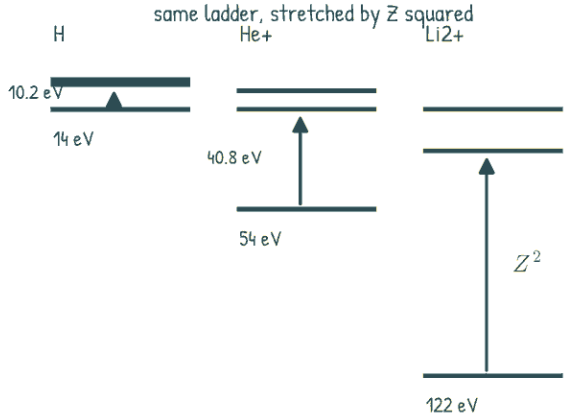
scaled ground states

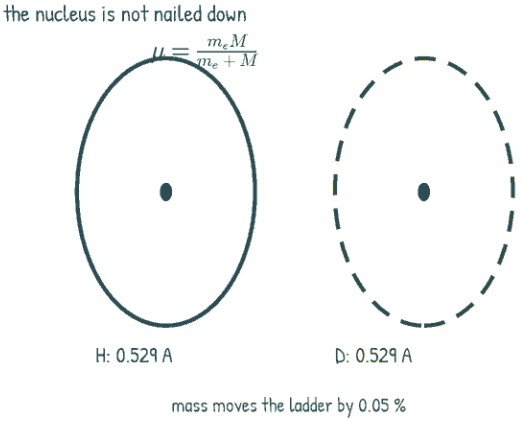
H −13.6 eV, 0.529 Å

He^+ −54.4 eV, 0.265 Å

Li^{2+} −122.4 eV, 0.176 Å

Z squared deepens the ladder; the reduced mass shifts every rung by a part in 10^4





KEEP / charge scales everything by Z², mass by the reduced mass — nothing else changes

▼1.89 eV
0.1 eV 10 MeV

What are the other three numbers for?

? Bohr gave one integer. Spectroscopy needs four, and each one explains a different splitting.

1 The numbers and their counts

For a given n, l runs 0…n−1, m_l runs −l…+l, and m_s is ±½. Counting states gives 2n² per shell, which is the periodic-table arithmetic in one line.

$$n:\ 1, 2,\ \ldots;\quad l < n;\quad -l \le m_l \le l;\quad m_s = \pm\tfrac{1}{2}$$

→ shell n holds 2n² states: 2, 8, 18, 32

2 A field turns one level into a fan

An orbiting electron is a current loop with magnetic moment μ_B per unit ħ. A field B splits a level into 2l+1 components spaced by μ_B B, which is why a line seen in a magnetic field becomes a triplet.

$$\Delta E = m_l \mu_B B, \qquad \mu_B = 5.79 \times 10^{-5}\ \mathrm{eV/T}$$

→ at 1 T the Hα line splits by 0.02 nm: small, but resolvable with a grating

3 Selection rules decide which lines exist

Not every pair of levels radiates: a photon carries one unit of angular momentum, so Δl = ±1 and Δm_l = 0, ±1. That is why some transitions are bright, others forbidden, and why the number of lines from a cascade is smaller than the number of level pairs.

$$\Delta l = \pm 1, \qquad \Delta m_l = 0,\ \pm 1$$

4 A worked cascade count

An atom in n = 5 cascades to the ground state. How many spectral lines, and how many from n = 5 alone?

$$N = \frac{5 \cdot 4}{2} = 10 \text{ lines}$$

$$\text{from } n = 5 \text{ only}:\ 5 \to 4, 3, 2, 1 = 4$$

→ the series count is n − n_f; the cascade count is all pairs

KEY RESULT
counts and splits

$$N = n(n-1)/2$$

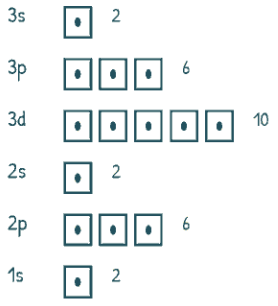
$$\Delta E = m_l \mu_B B$$

2n² states in shell n

GATE
what this model ignores
Spin-orbit fine structure (10^{-4} eV) is named, not derived.
The anomalous factor g ≈ 2 is outside the syllabus.

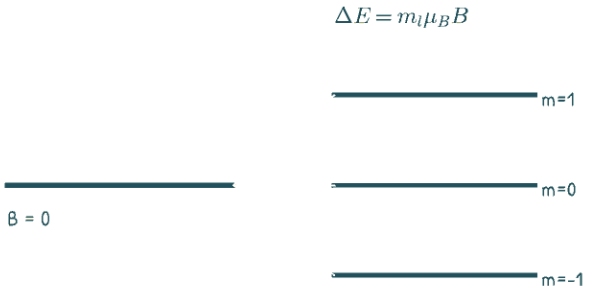
TRAP
reading the question
If it says "orbital only", use m_lμ_BB and do not double for spin.

The Bohr ladder is one quantum number; the other three decide which lines exist



orbitals e⁻ slots

n, l, m_l, m_s: 2n^2 states in shell n



a magnetic field fans one level into three

KEEP / n sets the energy; l, m_l, m_s decide which transitions are allowed and how they split

What exactly is in a beam of light?

? Counting coins: energy per photon, photons per second, and the push they give.

1

Energy per photon

One quantum carries hf, which in the units of this chapter is 1240 divided by the wavelength in nanometres. That single conversion does most of the numerical work in the chapter.

$$E = hf = \frac{hc}{\lambda} = \frac{1240}{\lambda\ [\text{nm}]}\ \text{eV}$$

→ green light at 550 nm: 2.25 eV; a 100 keV X-ray: 0.0124 nm

2

Photons per second, and per cubic metre

Divide the beam power by the energy per photon. Spread over a sphere, the intensity falls as 1/r², so the flux does too; inside the beam the number density is the flux divided by c.

$$\dot{N} = \frac{P}{hf}, \qquad n = \frac{I}{hf}, \qquad n_{\text{vol}} = \frac{I}{hfc}$$

→ a 100 W sodium lamp at 589 nm emits 3×10²⁰ photons per second

3

A beam pushes

Each photon arrives with momentum h/λ and leaves with either none (absorbed) or the reverse (reflected). The force is the momentum delivered per second, which is P/c or 2P/c. This is radiation pressure, derived here by counting rather than by fields.

$$F = \frac{P}{c}(1+R), \qquad p_{\text{rad}} = \frac{I}{c}\,(1+R)$$

→ the force from a 1 W reflected beam is 6.7 nN — a microgram's weight

4

A worked photon count

A 5 mW laser at 633 nm: how many photons leave per second, and what force does the beam exert if absorbed?

$$E = \frac{1240}{633} = 1.96\ \text{eV} = 3.14 \times 10^{-19}\ \text{J}$$

$$\dot{N} = \frac{5 \times 10^{-3}}{3.14 \times 10^{-19}} = 1.6 \times 10^{16}\ \text{s}^{-1}$$

→ F = P/c = 1.7×10⁻¹¹ N — a piconewton, and optical tweezers use exactly this

KEY RESULT

the three lines

$$E = 1240/\lambda\ (\text{eV, nm})$$

$$p = h/\lambda$$

$$F = P(1+R)/c$$

TRAP

massive-photon thinking

Using p = E/c for an electron. That relation is for massless particles; electrons take p = √(2mK).

Forgetting the factor 2 when a surface reflects.

NUMBERS

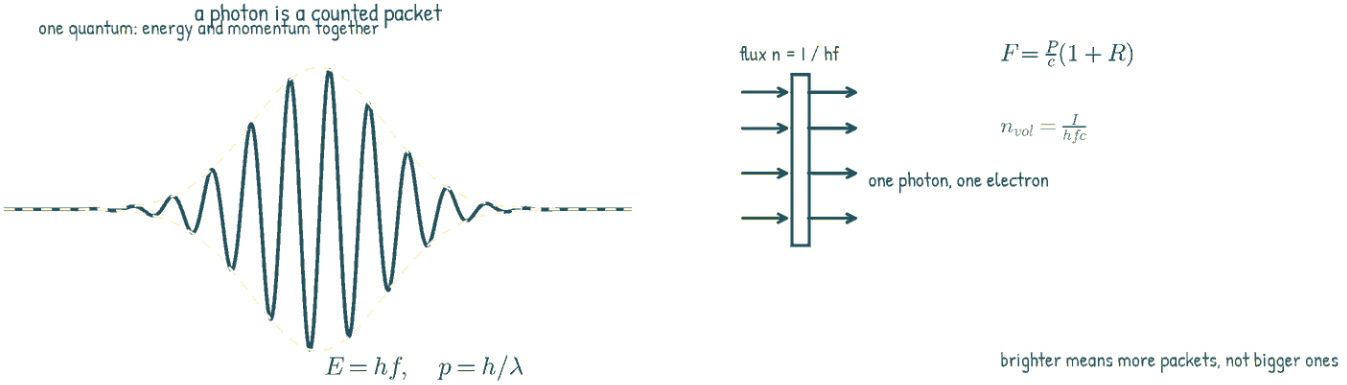
one-line conversions

hc = 1240 eV nm = 1.99×10⁻²⁵ J m

1 eV = 1.602×10⁻¹⁹ J

h = 6.63×10⁻³⁴ J s

Photon bookkeeping: energy per packet, and packets per second



KEEP / a beam is n photons per second, each carrying hf and momentum h/λ

▼1.8 eV
0.1 eV 10 MeV

Why does brighter light not give faster electrons?

? Five experimental facts, and the two that no wave theory can survive.

1 The five facts

There is a threshold frequency below which nothing is emitted at any intensity. Emission is instantaneous, within about a nanosecond. Current is proportional to intensity. Maximum kinetic energy grows with frequency and ignores intensity. The stopping potential is therefore a function of frequency alone.

→ three of the five are also what a wave predicts — they are decoys, not discriminators

2 Einstein's one line

One photon is absorbed by one electron. The electron pays a toll φ to escape the surface and keeps the rest as kinetic energy. Nothing about intensity appears here, which is why the maximum energy ignores it.

$$K_{\max} = hf - \phi, \qquad eV_s = K_{\max}$$

→ if hf < φ the right-hand side is negative and the answer is "no emission"

3 The graph that measured Planck's constant

Plot stopping potential against frequency. Every metal gives the same slope h/e and a different intercept −φ/e. Millikan's measurements of that slope agreed with the blackbody value of h, which is the cross-check that made the photon unavoidable.

$$V_s = \frac{h}{e}f - \frac{\phi}{e}, \qquad f_0 = \frac{\phi}{h}$$

→ slope 4.14×10^{-15} V s for every metal ever measured

4 A worked zinc threshold

Zinc has φ = 4.3 eV. What is the longest wavelength that ejects electrons, and what happens at 250 nm?

$$\lambda_0 = \frac{1240}{4.3} = 288 \text{ nm}$$

$$250 \text{ nm}: \; K_{\max} = \frac{1240}{250} - 4.3 = 4.96 - 4.3 = 0.66 \text{ eV}$$

→ visible light (400–700 nm) never discharges zinc: the photons are simply too poor

KEY RESULT
the balance

$$K_{\max} = hf - \phi$$

$$eV_s = hf - \phi$$

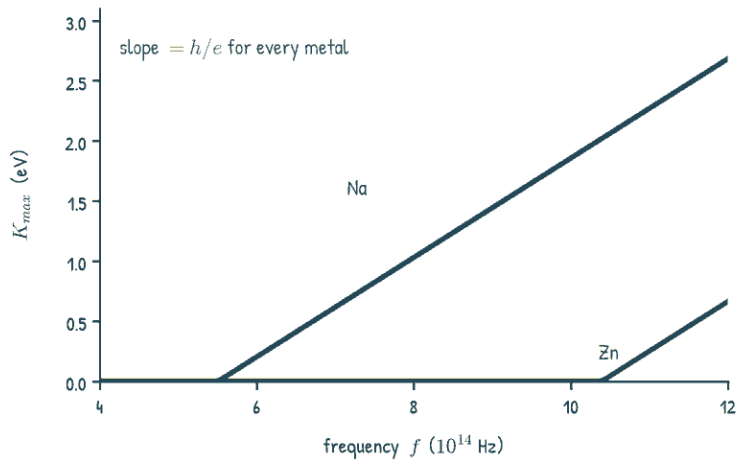
threshold $\lambda_0 = 1240/\varphi$ (nm, eV)

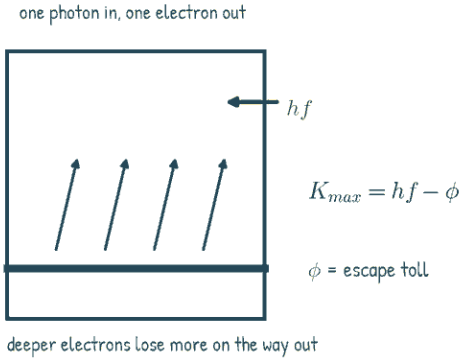
TRAP
four automatic zeros
Brighter light raising V_s.
Writing K_max = hf and forgetting the toll.
Expecting emission at threshold with energy: at λ_0 electrons emerge with zero.

NUMBERS
work functions
Na 2.28 eV · Zn 4.3 eV · Cs 1.9 eV · Pt 5.6 eV

Einstein equation: the photon pays a toll, the rest becomes kinetic energy

K_max = hf - phi





KEEP / intensity sets the number of electrons; frequency sets their maximum energy

How do you read a photocurrent graph?

? Two plateaux, one cutoff. Learn which knob moves which feature.

1 One curve, three features

Sweep the collector voltage. At large negative voltage the current is zero — even the fastest electron is turned back. Raise the voltage, electrons are collected, and at large positive voltage every emitted electron is collected: the current saturates.

$$eV_s = K_{\max}, \qquad I_{\text{sat}} \propto \text{ emission rate}$$

→ the saturation plateau is a collection limit, not a quantum effect

2 Doubling the intensity

Twice the photons means twice the electrons, so the plateau doubles. But each electron still gets the same photon energy, so the cutoff stays exactly where it was. Two curves, one edge.

→ any answer that moves the cutoff with intensity contradicts the whole experiment

3 Raising the frequency instead

At fixed intensity, higher frequency means fewer photons per second — each carries more energy — so the current actually falls slightly while the cutoff climbs. This is the subtle row that examiners use to separate people who track photons from people who track watts.

$$n = \tfrac{I}{hf}: \quad f\uparrow \;\Rightarrow\; n\downarrow \text{ at fixed } I$$

4 A worked graph reading

A V_s against f line passes through (5.0×10¹⁴ Hz, 0) and (11.0×10¹⁴ Hz, 2.48 V).

$$\text{slope} = \frac{2.48}{6.0\times10^{14}} = 4.13\times10^{-15}\ \text{V s}$$

$$h = e(4.13\times10^{-15}) = 6.62\times10^{-34}\ \text{J s}, \quad \phi = hf_0 = 2.07\ \text{eV}$$

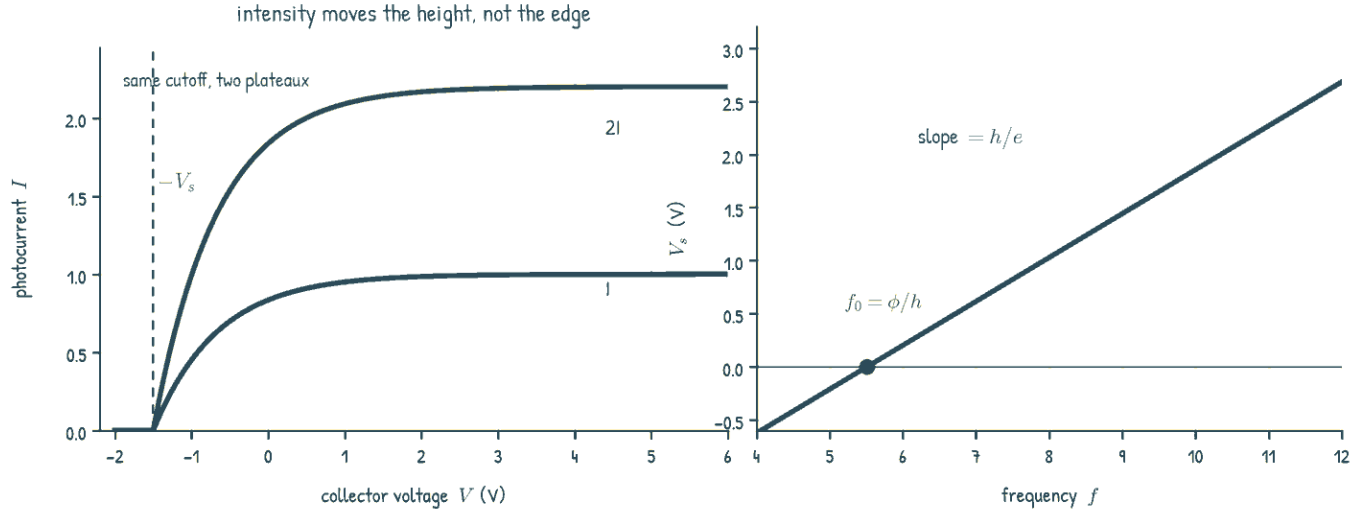
→ the slope must be h/e = 4.14×10⁻¹⁵ V s for every metal ever measured

KEY RESULT
read the graph
$eV_s = K_{\max}$
Left intercept: −V_s. Plateau height: rate. Slope of V_s vs f: h/e.

GATE
assumptions
One photon per electron; clean surface with a single φ; no field emission; no space charge.

TRAP
saturation is not quantum
Citing the saturation current as evidence for photons. It is geometry — every emitted electron collected.

Plot V_s against f: slope h/e, intercept −phi/e – Millikan measured h this way



KEEP / the cutoff marks the fastest electron; the plateau marks how many there are

What does a photocell measure, and what does it not?

? A battery and an ammeter separate two quantities that classical physics could not.

1 The apparatus is a filter

An evacuated tube holds an emitter and a collector. Light ejects electrons from the emitter; a variable voltage either helps them across or turns them back. The ammeter reads how many arrive, the voltmeter tells you at what retarding voltage they stop.

→ evacuated matters: in air the electrons collide with molecules within micrometres

2 Three knobs, three different answers

Intensity, frequency and collector voltage each move one feature and leave the others alone. That independence is the experimental content of the photon model, and it is why the photocell is a measuring instrument rather than a demonstration.

$$I_{\text{sat}} \propto I, \qquad V_s = \frac{hf - \phi}{e}$$

3 Where the ideal picture leaks

Real photocells have a spread of work functions, so the knee is soft rather than sharp. Very high intensity brings space-charge limiting; very high frequency makes the window absorb. None of these changes the threshold logic; they change the shape of the knee.

→ a soft knee is not evidence against quantisation — it is a dirty surface

4 A worked quantum efficiency

A photocell at 400 nm receives 1 mW and delivers 20 µA.

$$\dot{N}_\gamma = \frac{10^{-3}}{4.97 \times 10^{-19}} = 2.0 \times 10^{15}\ \text{s}^{-1}$$

$$\dot{N}_e = 1.25 \times 10^{14}\ \text{s}^{-1} \ \Rightarrow\ \eta = 6.2\ \%$$

→ a few per cent is normal for a plain metal; caesiated surfaces reach tens of per cent

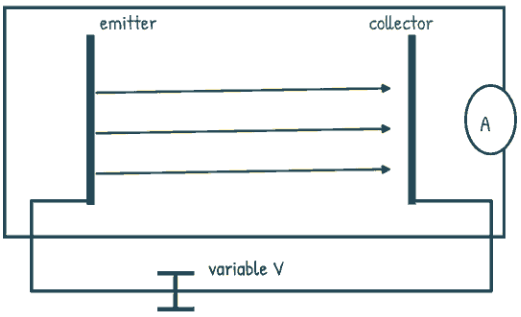
KEY RESULT
what each knob does
intensity → plateau
frequency → cutoff
collector voltage → collection only

INSIGHT
why it matters
The same device is a light meter, a night-vision tube and a solar cell's front end: one photon in, one electron out.

TRAP
current versus frequency
At fixed intensity, higher f gives fewer photons and a slightly lower current — not a higher one.

The photocell separates rate (intensity) from energy (frequency) with one battery

one battery measures both cutoff and rate

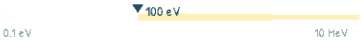


knob	current	V_s
intensity up	up	same
frequency up	down a little	up
voltage up	saturates	same

fixed I, higher f: fewer, richer photons

so the current dips while the cutoff climbs

KEEP / the battery measures energy per electron; the ammeter measures electrons per second



Do electrons have a wavelength?

? If light can be a particle, matter can be a wave. The only question is how big.

1 The hypothesis, and its one-line consequence

Attach to any particle the same relation that photons obey. For an electron accelerated from rest through V volts, the kinetic energy fixes the momentum, and the wavelength follows.

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_e eV}} = \frac{1.226}{\sqrt{V}}\ \text{nm}$$

→ 100 V gives 0.123 nm — the spacing of atoms, which is why crystals diffract electrons

2 Why you never see a cricket ball diffract

A 0.15 kg ball at 30 m/s has $\lambda \approx 10^{-34}$ m. Wave behaviour appears only when the wavelength is comparable to the geometry it meets, so a ball needs slits 10^{-34} m apart. The classical world is the limit $\lambda \ll$ every dimension.

$$\lambda_{\text{ball}} = \frac{6.63 \times 10^{-34}}{(0.15)(30)} = 1.5 \times 10^{-34}\ \text{m}$$

3 Electron microscopes are the practical payoff

Resolution is limited by the wavelength used. At 10 kV an electron has λ = 0.012 nm, four orders finer than green light, so the same lens geometry can in principle resolve viral structure. Aberrations eat most of that gain; they do not remove it.

→ slow electrons (tens of eV) have wavelengths of a few ångström — ideal for surfaces

4 A worked wavelength pair

Electrons at 100 V and at 10 kV:

$$\lambda_{100} = \frac{1.226}{\sqrt{100}} = 0.123\ \text{nm}$$

$$\lambda_{10^4} = 0.0123\ \text{nm}$$

→ a hundredfold voltage gives exactly a tenfold shorter wave: $\lambda \propto V^{-1/2}$

KEY RESULT

the bridge

$$\lambda = h/p$$

$$\lambda = 1.226/\sqrt{V}\ \text{nm}$$

p = √(2mK) non-relativistic; p = √(K² + 2mKc²)/c exact

TRAP

the four percent

The square-root formula is wrong above about 3 kV by more than 1 %.

And p = E/c is only for massless particles.

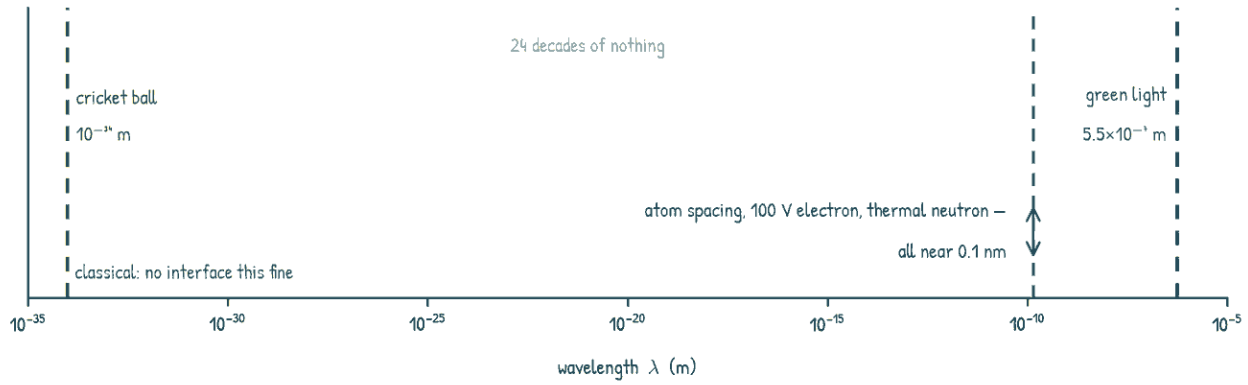
NUMBERS

bench wavelengths

100 V: 0.123 nm · 10 kV: 0.0123 nm

thermal neutron: 0.18 nm · green light: 550 nm

The same formula covers a cricket ball and an electron; only the ratio to the slit decides



KEEP / λ = h/p: large for electrons at a few volts, absurdly small for anything you can hold

Was the electron wave ever actually seen?

? Two experiments in 1927 turned a hypothesis into a measurement: a crystal and a detector.

1

A crystal is a grating with ångström spacing

Atoms in a crystal sit 1–3 Å apart, which is the wavelength of a 50–150 V electron. The crystal therefore acts as a three-dimensional grating, and reflected beams appear only at angles where the path difference between successive planes is a whole number of wavelengths.

$$2d\sin\theta = n\lambda$$

→ θ is the glancing angle, measured from the plane — not from the normal

2

The 1927 numbers

Davisson and Germer fired 54 V electrons at nickel and found a strong lobe at 65° from the incident beam. With the known plane spacing of 0.91 Å, Bragg predicts a wavelength of 1.65 Å; de Broglie gives 1.67 Å. The two agree to the accuracy of the day.

$$\lambda_{\text{Bragg}} = 2(0.91)\sin 65^\circ = 1.65\ \text{Å}$$

$$\lambda_{\text{deB}} = \frac{1.226}{\sqrt{54}} = 1.67\ \text{Å}$$

3

What followed

The same logic runs today as LEED in every surface lab, and as electron diffraction in structural biology. Slow electrons are used deliberately: a long wavelength both diffracts strongly and penetrates only a few atomic layers, so the pattern is a picture of the surface.

→ slower electrons diffract more, not less — lower energy means longer wavelength

4

A worked angle

54 V electrons on nickel planes of 0.91 Å:

$$\lambda = 1.67\ \text{Å}, \quad \sin\theta = \frac{1.67}{1.82} = 0.918$$

$$\theta = 66^\circ \quad \text{observed } 65^\circ$$

→ the one-degree gap is the rounding of d in the textbook value, not a failure of the theory

KEY RESULT

the pair

$$2d\sin\theta = n\lambda$$

$$\lambda = h/p$$

Bragg gives the wavelength from geometry; de Broglie predicts it from voltage.

GATE

why the peak is sharp

Constructive interference over thousands of planes; a small angle error destroys the sum.

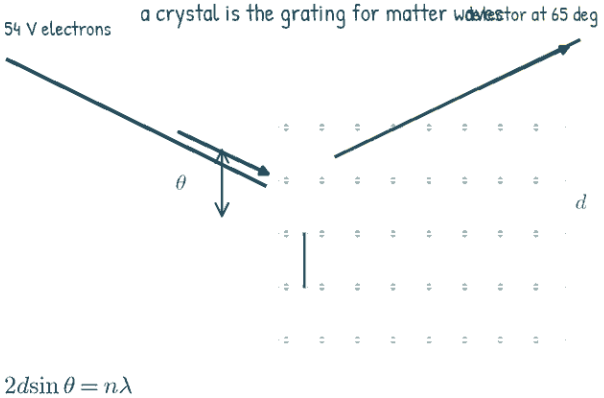
TRAP

angle and order

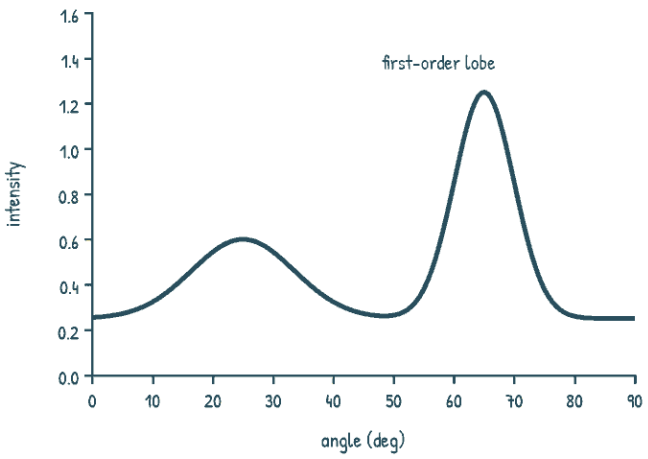
Writing 2d cosθ, or calling the second observed peak n = 1.

Forgetting that the detector angle is 2θ.

Davisson-Germer: the 65 degree lobe sits where 2d sin(theta) = lambda predicts



$2d\sin\theta = n\lambda$



KEEP / the diffraction angle follows 2d sinθ = nλ, and λ is exactly h/p

▼13.6 eV
0.1 eV 10 MeV

Why does the electron not fall into the nucleus?

? A classical orbiting charge radiates and dies in picoseconds. Something holds the atom up.

1 A localised particle cannot have a definite momentum

A pure sine has a perfect wavelength and no position. To localise a particle you superpose many waves; the narrower the packet, the wider the spread of momenta.

$$\Delta x\,\Delta p \gtrsim \frac{\hbar}{2}$$

→ a statement about waves, not about clumsy equipment; squeeze further and the kinetic term wins

2 Minimise the total energy

Confine an electron to radius r: its momentum is at least $\hbar/r$, so the kinetic term grows as $1/r^2$ while the Coulomb well deepens as $-1/r$. The sum has a minimum, and that minimum is the hydrogen atom.

$$E(r) \approx \frac{\hbar^2}{2m_e r^2} - \frac{ke^2}{r}$$

$$r_{\min} = \frac{\hbar^2}{m_e k e^2} = 0.529\ \text{Å}, \qquad E_{\min} = -13.6\ \text{eV}$$

3 Two more payoffs you will use

A particle in a small box keeps a minimum energy even at zero temperature — why helium stays liquid and nucleons stay energetic. And a state living only τ seconds has an energy width $\hbar/\tau$, the natural width of every spectral line.

$$K_{\min} \approx \frac{\hbar^2}{2mL^2}, \qquad \Delta E \sim \frac{\hbar}{\tau}$$

4 A worked ground state

Evaluate the two terms of E(r) at r = 0.5 Å:

$$K = \frac{\hbar^2}{2m_e r^2} = 15.3\ \text{eV}, \qquad U = -\frac{ke^2}{r} = -28.8\ \text{eV}$$

$$E = 15.3 - 28.8 = -13.5\ \text{eV}$$

→ the exact answer is −13.6 eV: the estimate misses by one per cent, and the physics is all there

KEY RESULT
the three uses
$\Delta x \Delta p \gtrsim \hbar/2$

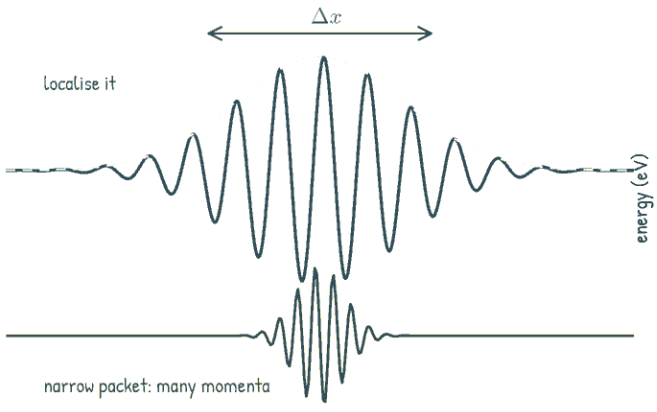
$E(r) = \hbar^2/2mr^2 - ke^2/r$

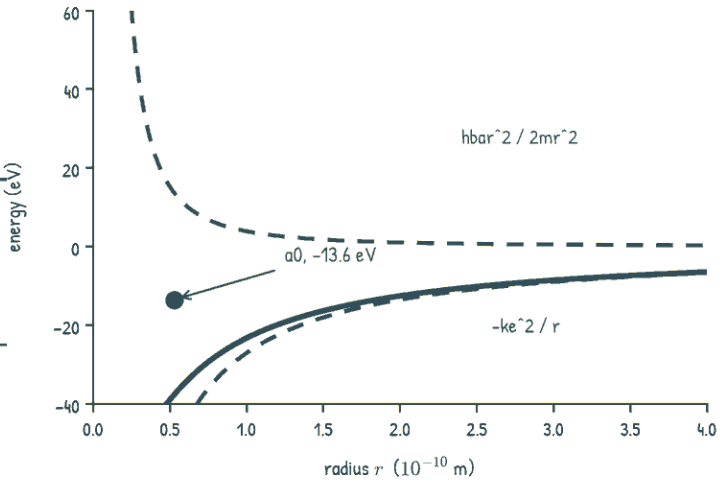
Confinement, zero-point energy, natural linewidth.

INSIGHT
why it is exact here
For the hydrogen ground state the estimate saturates the bound, so it returns a_0 and −13.6 eV exactly. Do not expect that for every state.

TRAP
the missing half
Writing $\Delta x \Delta p \geq \hbar$, or L = nh. The bound is $\hbar/2$ and L = $n\hbar$.

Squeezing costs kinetic energy: the sum has a floor, and that floor is the atom





KEEP / confinement costs momentum; the ground state is where that cost stops paying

0.1 eV 1 MeV 10 MeV

Why does every nucleus have the same density?

? Add nucleons and the ball grows, but the stuff inside never gets denser. What fixes the size?

1 A saturated force makes a fixed-volume ball

The strong force is attractive at 1–2 fm, gone by about 3 fm, and charge-independent. Because it is short-range, each nucleon binds only its neighbours: doubling A doubles the binding, not the binding per nucleon. Constant energy per nucleon means constant volume per nucleon.

$$V = Av = \tfrac{4}{3}\pi R^3 \;\Rightarrow\; R = R_0 A^{1/3}, \qquad R_0 = 1.2 \text{ fm}$$

→ the cube root is the whole content: density, not radius, is the invariant

2 Read the density off the constant

Substituting the fitted radius gives a density with no A in it. That is why the answer to "carbon or uranium?" is "neither, they are the same" — a fact worth carrying into any estimate.

$$\rho = \frac{3m_N}{4\pi R_0^3} = 2.3 \times 10^{17} \text{ kg/m}^3$$

→ a 5 cm³ teaspoon of it weighs 1.2×10¹² kg — a mountain in a spoon

3 What the numbers actually are

Carbon-12 has R = 2.75 fm, uranium-238 has R = 7.44 fm: only 2.7 times bigger for twenty times the nucleons. The nucleus occupies 10^{-12} of the atom's volume, yet holds 99.97 % of its mass.

$$R_{12} = 1.2(12)^{1/3} = 2.75 \text{ fm}, \qquad R_{238} = 1.2(238)^{1/3} = 7.44 \text{ fm}$$

→ the electron cloud is 10^5 times wider, and almost all of the atom is empty

4 A worked uranium radius

Uranium-238: radius and the mass of a 5 cm³ spoonful.

$$R = 1.2(238)^{1/3} = 7.44 \text{ fm}, \qquad V = \tfrac{4}{3}\pi R^3 = 1.72 \times 10^{-42} \text{ m}^3$$

$$m = \rho V = (2.3 \times 10^{17})(5 \times 10^{-6}) = 1.2 \times 10^{12} \text{ kg}$$

→ a billion tonnes in a teaspoon — the number that makes the density believable

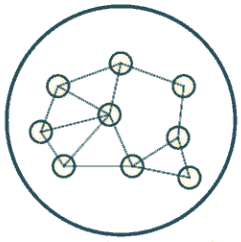
KEY RESULT
size and density
$R = 1.2A^{1/3}$ fm
$\rho = 2.3 \times 10^{17}$ kg/m³
Same for every nucleus.

GATE
where the law is soft
Liquid-drop mean radius. Halo nuclei and the surface fuzz bend it at the few-percent level.
It says nothing about shape: deformed nuclei have two radii.

TRAP
the two confusions
Saying density falls as A grows (it does not), and taking R ∝ A instead of A^1/3.

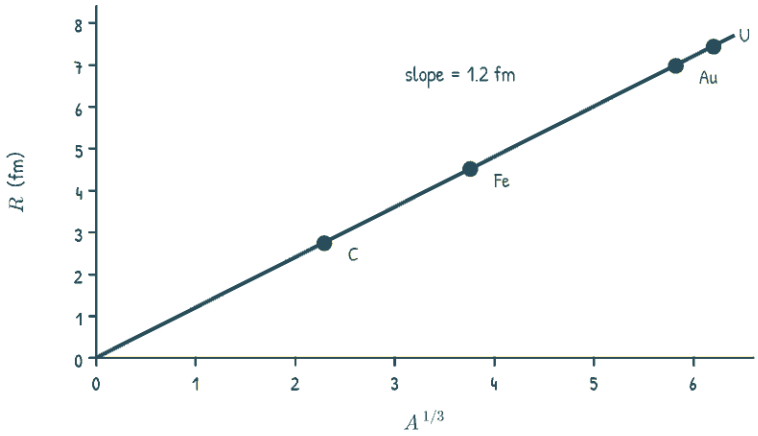
Saturated binding gives every nucleon the same volume: R = 1.2 A^(1/3) fm, density fixed

saturated attraction binds neighbours only



Coulomb repulsion: every proton pair, no saturation

radius grows as the cube root of A



KEEP / R = 1.2 A^1/3 fm is the statement that nuclear matter is incompressible

0.1 eV 100 MeV 10 MeV

Where does the missing mass go?

? Weigh a nucleus and its parts: the whole is lighter. The difference is what holds it together.

1

Weigh the parts, weigh the whole

Take Z protons and N neutrons apart, then assemble them. The assembly is lighter by Δm, and that mass went out as the energy that bound it. Conversely, Δm c² is the energy you must pay to take it apart again.

$$B = [Zm_H + Nm_n - M_{\text{atom}}]c^2, \qquad 1\ \text{u} = 931.5\ \text{MeV}$$

→ atomic masses are used throughout: the electron bookkeeping cancels for the nucleus

2

Two numbers set the scale

The deuteron is barely bound: 2.22 MeV for two nucleons, 1.11 each. Helium-4 is the first tight package: 28.3 MeV, 7.07 per nucleon — a factor 6.4 jump that explains why fusing past hydrogen pays so much.

$$\Delta m_d = 0.002388\ \text{u} \Rightarrow B = 2.22\ \text{MeV}, \qquad B/A = 1.11$$

$$\Delta m_{He} = 0.030377\ \text{u} \Rightarrow B = 28.3\ \text{MeV}, \qquad B/A = 7.07$$

3

The curve is the decision-maker

Plot B/A against A and every energy question becomes "which way does this move the nucleons?". Fusion climbs the left slope toward iron; fission climbs the right slope toward the same summit. Anything that moves nucleons up the curve releases the difference.

$$\text{fission}: \ 7.6 \to 8.5\ \text{MeV/nucleon} \ \Rightarrow \ \sim 200\ \text{MeV}$$

→ iron has nothing to gain — hence the endpoint of stellar burning

4

A worked mass defect

Helium-4 from its parts: 2 hydrogen atoms and 2 neutrons.

$$\Delta m = 2(1.007825) + 2(1.008665) - 4.002603 = 0.030377\ \text{u}$$

$$B = 0.030377(931.5) = 28.30\ \text{MeV}, \qquad B/A = 7.07\ \text{MeV}$$

→ the deuteron gives 1.11 per nucleon: that factor 6.4 is why fusion past hydrogen pays

KEY RESULT

the two lines

$$B = \Delta m\, c^2$$

$$1\ \text{u} = 931.5\ \text{MeV}$$

B/A: 1.1 (d) → 7.1 (He) → 8.8 (Fe) → 7.6 (U).

INSIGHT

why the peak

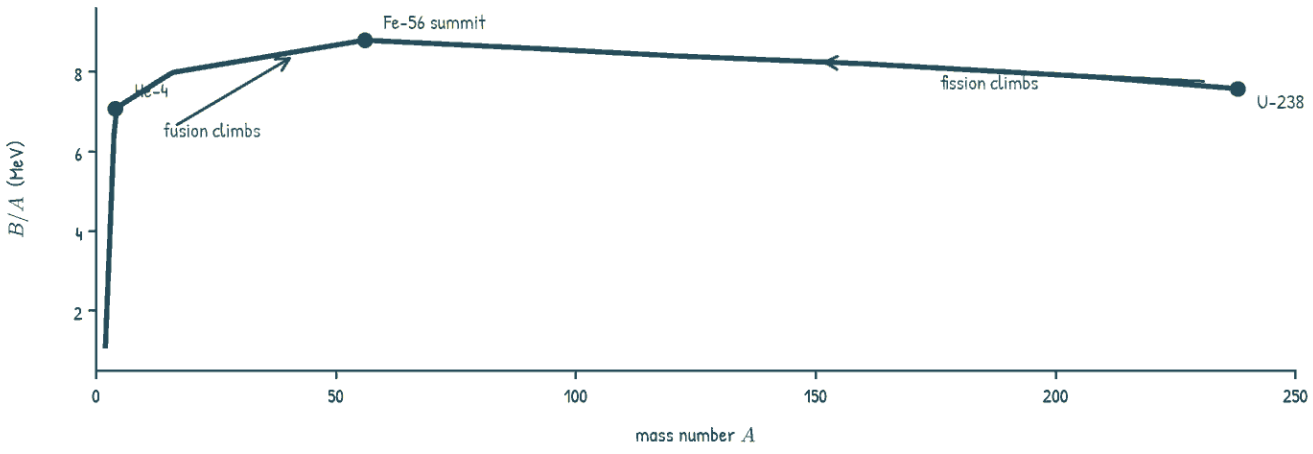
Short-range attraction grows as A, Coulomb repulsion as Z². Below iron the surface tax dominates; above it the Coulomb tax does.

TRAP

the sign

Using bare proton masses instead of hydrogen atoms in a beta problem, or calling the mass defect a loss of nucleons.

Any process that moves nucleons up the B/A curve releases the difference; iron is the summit



KEEP / B = Δm c², 1 u = 931.5 MeV, and B/A peaks at iron near 8.8 MeV

Why does the valley of stability bend?

? Light nuclei want equal protons and neutrons. Heavy ones want extra neutrons. The liquid drop says why.

1 Five terms, five physical reasons

Volume buys binding for every nucleon; surface taxes the ones at the edge; Coulomb punishes every proton pair; asymmetry punishes N ≠ Z because the two Fermi seas must be filled to the same height; pairing rewards even numbers.

$$B = a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_a \frac{(N-Z)^2}{A} + \delta$$

→ typical coefficients 15.8, 18.3, 0.71, 23.2, 12 MeV

2 Maximise over Z at fixed A

Only the Coulomb and asymmetry terms care about Z once A is fixed. Setting the derivative to zero gives the valley floor: the most stable Z for that mass number.

$$Z_0 = \frac{A}{2 + \frac{a_c}{2a_a} A^{2/3}} = \frac{A}{2 + 0.0154 A^{2/3}}$$

→ A = 100 gives 42.9; A = 238 gives 92.0 — uranium sits on its own floor

3 Off the floor, beta decay is the traffic

A nucleus with too many neutrons converts one into a proton; too few, the reverse. The valley is the map and beta decay is the traffic that moves every nucleus toward the floor.

$$\beta^- : n \to p + e^- + \bar{\nu}_e \qquad \beta^+ : p \to n + e^+ + \nu_e$$

→ A never changes in beta decay; only Z does

4 A worked valley floor

Most stable charge for A = 100 and A = 238.

$$Z_0(100) = \frac{100}{2 + 0.0154(100)^{2/3}} = \frac{100}{2.33} = 42.9$$

$$Z_0(238) = \frac{238}{2 + 0.0154(238)^{2/3}} = 92.0$$

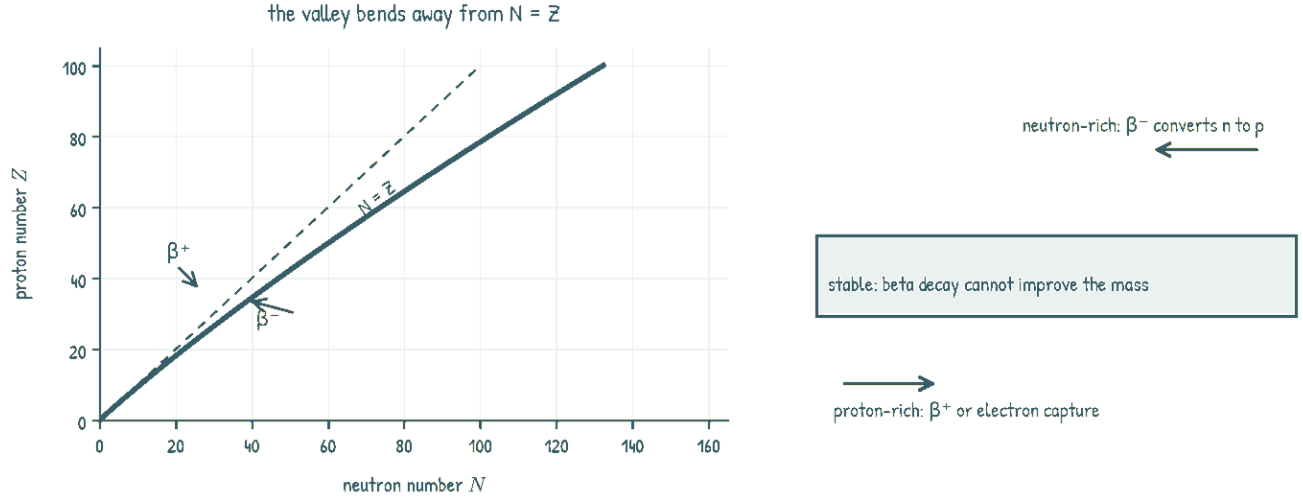
→ molybdenum-ruthenium for the first, uranium itself for the second

KEY RESULT
the floor
$$Z_0 = \frac{A}{2 + 0.0154 A^{2/3}}$$
Light nuclei: N ≈ Z. Heavy: N/Z ≈ 1.5.

GATE
liquid drop only
Shell closures dent and bump the floor — the model is the trend, not the detail.
Beyond bismuth no configuration is fully stable.

TRAP
which way
Neutron-rich nuclei go β^- (Z up). Proton-rich go β^+ or EC (Z down). Getting this backwards is the most common slip.

Coulomb grows as Z², surface as A^(2/3), so heavy nuclei need extra neutrons



KEEP / Coulomb grows as Z², surface as A^2/3, so Z0 = A/(2 + 0.0154 A^2/3)

Why does the alpha carry almost all the energy?

? Two bodies leave a decay at rest. Momentum must cancel, so the light one takes the lion's share.

1 Q is one subtraction

For any decay, Q = (mass in − mass out)c². Positive Q means it can happen; it says nothing about how fast. With atomic masses the electrons cancel, so the arithmetic is one line.

$$Q = (M_{\text{parent}} - M_{\text{daughter}} - M_{\alpha})c^2$$

→ alpha decay of U-238: Q = 4.27 MeV, which is why it can happen at all

2 Momentum splits Q inversely to mass

The parent was at rest, so the two products leave back to back with equal momentum. Kinetic energy goes as p²/2m, so the lighter body takes the larger share — the same rule as a recoiling gun.

$$K_{\alpha} = \frac{A-4}{A}Q, \qquad K_{\text{daughter}} = \frac{4}{A}Q$$

→ U-238: 4.20 MeV to the alpha, 0.07 MeV to the recoiling thorium

3 Discrete energies, because levels are discrete

The alpha can leave the daughter in its ground state or in an excited one; each branch has its own Q, so the alpha spectrum is a set of sharp lines, not a continuum. Any leftover excitation leaves later as a gamma.

$$Q_{\text{branch}} = Q_0 - E_{\text{ex}}, \qquad E_{\gamma} = E_{\text{ex}}$$

→ this is the nuclear analogue of an atomic line spectrum

4 A worked alpha split

Polonium-210 decays with Q = 5.40 MeV.

$$K_{\alpha} = \frac{206}{210}(5.40) = 5.30 \text{ MeV}, \qquad K_d = 0.10 \text{ MeV}$$

$$\sqrt{2m_{\alpha}K_{\alpha}} = \sqrt{2m_dK_d}$$

→ the daughter recoils with 2 % of the energy and a hundred times less speed

KEY RESULT
the split

$$Q = (m_{\text{in}} - m_{\text{out}})c^2$$

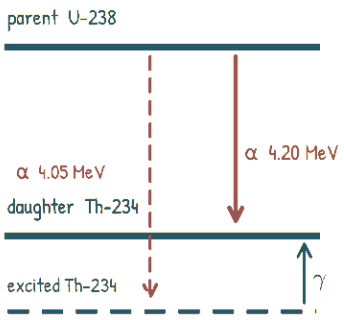
$$K_{\alpha} = \frac{A-4}{A}Q$$

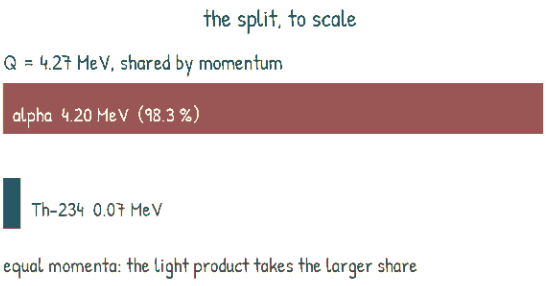
Lighter product takes the larger share.

GATE
two bodies only
The formula assumes the parent was at rest and only two products leave.
Beta decay is three bodies — hence its continuous spectrum.

TRAP
what Q is not
Q is not the alpha kinetic energy (multiply by (A−4)/A), and a positive Q does not mean an instant decay.

Q from masses; momentum splits it as (A−4)/A to the alpha and 4/A to the daughter





KEEP / Q from masses, then K_alpha = (A−4)Q/A — momentum fixes the split

0.1 eV 1 MeV 10 MeV

How can a beta spectrum be continuous?

? Alpha lines are sharp. Beta electrons arrive with every energy from zero to an endpoint. Something is missing.

1 Three bodies, not two

In beta decay the nucleus emits an electron and an antineutrino. With three bodies leaving, the energy can be shared in a continuum of ways: every electron energy from zero up to the endpoint Q is allowed, and the neutrino takes the remainder.

$$n \to p + e^- + \bar{\nu}_e \qquad p \to n + e^+ + \nu_e$$

→ the endpoint E_0 = Q is where the neutrino gets nothing

2 The electron rule, in one line each

Atomic masses include the electrons, and that changes the bookkeeping between the two beta signs. For β^- the daughter has one more electron, which the atomic mass already counts. For β^+ the positron is an extra particle and the atom has one electron too few.

$$Q_{\beta^-} = (M_P - M_D)c^2, \qquad Q_{\beta^+} = (M_P - M_D - 2m_e)c^2$$

→ C-14: 0.156 MeV. N-13: 2.220 − 1.022 = 1.20 MeV

3 Gamma and conversion finish the job

The daughter often stays excited. It can emit a photon of the level gap (gamma), or hand the same energy to an orbital electron (internal conversion), which then leaves with the gap minus its binding. Both carry the same information: the nuclear level diagram.

$$E_\gamma = E_{\text{ex}}, \qquad K_e = E_{\text{ex}} - B_{\text{shell}}$$

→ long-lived excited states are isomers; they decay later with the same energy

4 Two worked endpoints

Carbon-14 and nitrogen-13, with the electron rule applied.

$$Q_{\beta^-} = (14.003242 - 14.003074)(931.5) = 0.156 \text{ MeV}$$

$$Q_{\beta^+} = (13.005739 - 13.003355 - 0.001098)(931.5) = 1.20 \text{ MeV}$$

→ the 2m_e term is 1.022 MeV — larger than most beta Q-values, which is why β^+ needs a big parent

KEY RESULT
the two rules

$$Q_{\beta^-} = (M_P - M_D)c^2$$

$$Q_{\beta^+} = (M_P - M_D - 2m_e)c^2$$

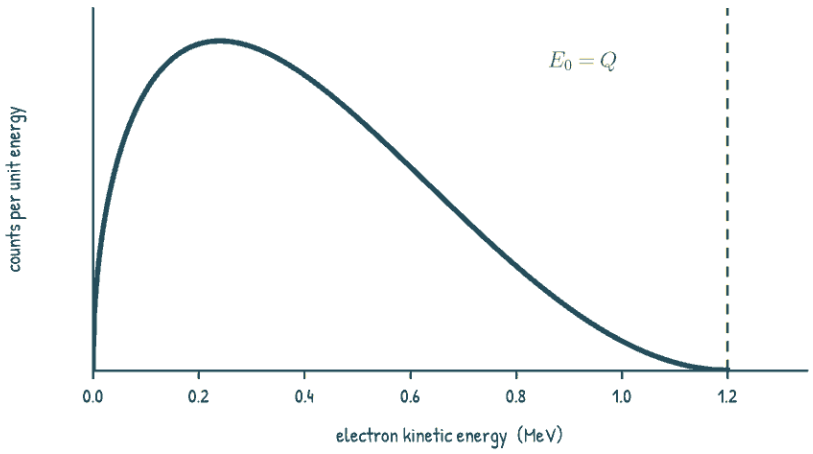
A is unchanged in both.

GATE
spectrum shape
The endpoint is exact; the shape near zero depends on the Coulomb pull on the emitted electron.
Neutrino mass is ~1 eV, invisible at MeV resolution.

TRAP
the classic trio
Giving the electron all of Q. Changing A by one. Forgetting the 2m_e for β^+.

A third particle carries the rest; the endpoint is where it carries nothing

N-13: every electron energy is allowed



two particles share Q



the gap between them is the missing energy

KEEP / a third particle carries the rest; the endpoint is where it carries nothing

0.1 eV 1 MeV 10 MeV

What does a half-life actually measure?

? Each nucleus has a fixed chance per second of dying. From that one assumption the whole calculus follows.

1 A memoryless probability gives an exponential

Assume every nucleus has the same chance λ dt of decaying in the next dt, however long it has already lived. Then dN = −λN dt integrates to an exponential, and the same λ governs the activity, because activity is λ times the number present.

$$N = N_0 e^{-\lambda t}, \qquad A_{ct} = \lambda N, \qquad T_{1/2} = \frac{\ln 2}{\lambda}$$

→ the log plot is a straight line of slope −λ; that is how half-lives are measured

2 Mean life is not half-life

The average lifetime is the integral of t over the decay distribution, and it comes out 1/λ — larger than the half-life by 1/ln2. The nucleus that survives longest is not the typical one.

$$\tau = \frac{1}{N_0}\int_0^{\infty} t\,(-dN) = \frac{1}{\lambda} = 1.443\,T_{1/2}$$

→ C-14: T½ 5730 y, τ 8270 y; Ra-226: 1600 y and 2310 y

3 Three things λ is not

λ does not depend on how many nuclei you have, on temperature, or on chemistry — with one exception: electron capture needs a bound electron, so extreme ionisation can shift it.

$$\lambda = \lambda_1 + \lambda_2 + \ \ldots\ , \qquad f_i = \frac{\lambda_i}{\lambda}$$

→ two decay modes add their constants; the branching fraction is the ratio

4 A worked working day

Tc-99m (T½ = 6 h) prepared at 800 MBq at 08:00; Co-60 mass for 1 Ci.

$$A(12\ \text{h}) = 800 \times 2^{-2} = 200\ \text{MBq}$$

$$N = \frac{3.7 \times 10^{10}}{4.17 \times 10^{-9}} = 8.9 \times 10^{18} \ \Rightarrow\ m = 0.88\ \text{mg}$$

→ a curie of a years-scale isotope is a speck; a curie of radium is a gram

KEY RESULT

the three

$$N = N_0 e^{-\lambda t}$$

$$A_{ct} = \lambda N$$

$$\tau = 1/\lambda = 1.443 T_{1/2}$$

INSIGHT

why λ is stable

Nuclear decay is a tunnelling or weak process at the MeV scale; room temperature is 25 meV. The ratio is 4×10⁷, so nothing ordinary can move it. Halving the sample halves A but leaves λ untouched.

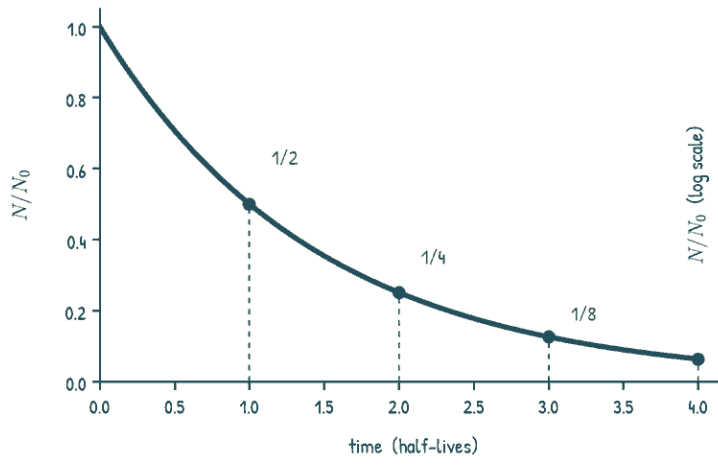
TRAP

the missing ln 2

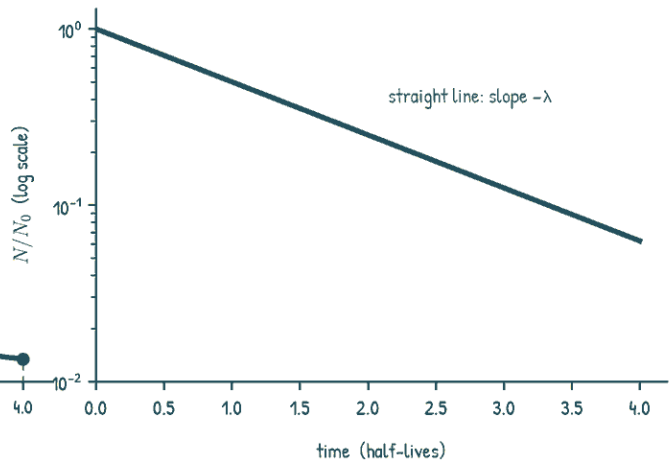
Writing e^{−t/T½} for e^{−λt}: at one half-life that gives 37 %, not 50 %. The exponent must carry ln 2.

N = N0 e^(−λt); the halving ladder for whole half-lives, the log line for the rest

each half-life halves the remainder



the log plot is the working tool



KEEP / N = N0e^{−λt}, T½ = ln2/λ, τ = 1.443 T½ — and λ is intensive

How do you date a bone?

? A living thing keeps its radiocarbon topped up. When it dies the clock starts, and the ratio is the hand.

1

The clock is an exchange that stops

Cosmic rays keep the atmospheric $^{14}C/^{12}C$ ratio roughly fixed. Living tissue exchanges carbon, so it matches the atmosphere; death stops the exchange and the ^{14}C already inside decays away.

$$\frac{A}{A_0} = e^{-\lambda t} \;\Rightarrow\; t = \frac{T_{1/2}}{\ln 2}\ln\frac{A_0}{A}$$

→ 10 disintegrations per minute per gram against 15 gives 3350 years

2

The range is set by counting, not by physics

After ten half-lives one nucleus in a thousand is left, and counting statistics dominate. That is why the ceiling is about 50 000 years, and why rocks need longer-lived clocks.

$$t_{10} = 10(5730) = 5.7\times10^{4}\ \text{y}, \qquad \frac{\sigma}{N} = \frac{1}{\sqrt{N}}$$

→ K-Ar: 1.25 Gyr; U-Pb: 4.47 Gyr — same logarithm, different constant

3

The assumptions you must state

Constant production, a closed sample, and a known initial ratio. The last is the weak link, so tree rings calibrate the raw ratio; the isochron method removes it by plotting one ratio against another.

$$\frac{^{87}\text{Sr}}{^{86}\text{Sr}} = \left(\frac{^{87}\text{Sr}}{^{86}\text{Sr}}\right)_0 + \frac{^{87}\text{Rb}}{^{86}\text{Sr}}\left(e^{\lambda t}-1\right)$$

→ a straight line whose slope is the age — the intercept absorbs the unknown start

4

A worked carbon age

A bone gives 10 disintegrations per minute per gram; living tissue gives 15.

$$t = \frac{5730}{\ln 2}\ln\frac{15}{10} = 8267(0.405) = 3350\ \text{y}$$

$$\frac{A}{A_0} = 2^{-t/T_{1/2}} = 0.667$$

→ between one and two half-lives, which is the range where the method is most reliable

KEY RESULT

the clock

$$t = \frac{T_{1/2}}{\ln 2}\ln\frac{A_0}{A}$$

C-14: 5730 y, range to ~50 ka.

K-Ar, U-Pb for rock.

GATE

when it fails

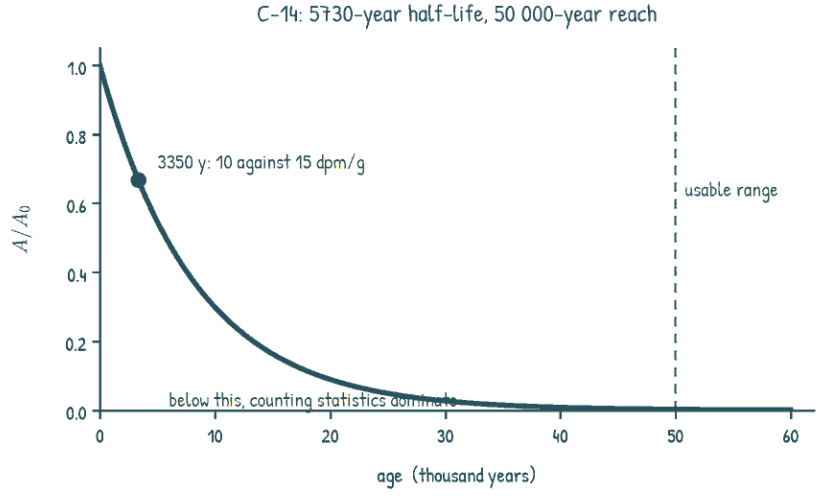
Open samples, contamination, fossil fuel (dead carbon), and dates near the limit where the count is a handful.

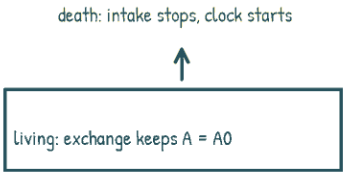
TRAP

the ratio inverted

Using A/A_0 instead of A_0/A flips the sign of the age. Check that a smaller activity gives a positive age.

The clock runs from the moment exchange stops; the range ends where the counts do





calibration curves correct the raw ratio

KEEP / $t = (T_{½}/\ln 2)\ln(A_0/A)$: three assumptions, one logarithm

0.1 eV 1 MeV 10 MeV

Why does a decay chain settle into a fixed ratio?

? A daughter fed by its parent has two clocks. When one is much slower, the other just follows.

1 Two clocks in one equation

The daughter is produced at rate $\lambda_1 N_1$ and destroyed at rate $\lambda_2 N_2$. Solving the pair with a daughter that starts at zero gives a rise-and-fall: the daughter activity climbs, peaks, then locks onto the parent's decay.

$$\frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2 \;\Rightarrow\; A_2 = \frac{\lambda_2}{\lambda_2 - \lambda_1} A_1 \left(1 - e^{-(\lambda_2 - \lambda_1)t}\right)$$

→ the bracket is the approach to equilibrium; after a few daughter half-lives it is one

2 Name the regime before doing arithmetic

If the parent is far longer-lived, the ratio settles at one: secular equilibrium, the state of old radium samples. If it is only somewhat longer-lived, the ratio settles at a number above one: transient equilibrium, and the chain decays with the parent's clock.

$$\text{secular}: \ \frac{A_2}{A_1} \to 1, \qquad \text{transient}: \ \frac{A_2}{A_1} \to \frac{\lambda_2}{\lambda_2 - \lambda_1}$$

→ $\lambda_2 = 3\lambda_1$ gives a ratio of 3/2; the daughter is always the more active one

3 Peak time and branching

The daughter activity peaks where its derivative vanishes; after that it inherits the parent's slope. If a nuclide has two decay modes, the constants add and each branch takes its share of the decays.

$$t_{\max} = \frac{\ln(\lambda_2/\lambda_1)}{\lambda_2 - \lambda_1}, \qquad \lambda = \lambda_a + \lambda_b, \quad f_a = \frac{\lambda_a}{\lambda}$$

→ the long-term tempo always belongs to the longest-lived member present

4 A worked generator

Parent $T_{1/2}$ = 18 h, daughter $T_{1/2}$ = 6 h: find the ratio and the peak.

$$\lambda_1 = 0.0385\ \text{h}^{-1}, \ \lambda_2 = 0.1155\ \text{h}^{-1}$$

$$\frac{A_2}{A_1} = \frac{0.1155}{0.0770} = 1.50, \qquad t_{\max} = \frac{\ln 3}{0.0770} = 14.3\ \text{h}$$

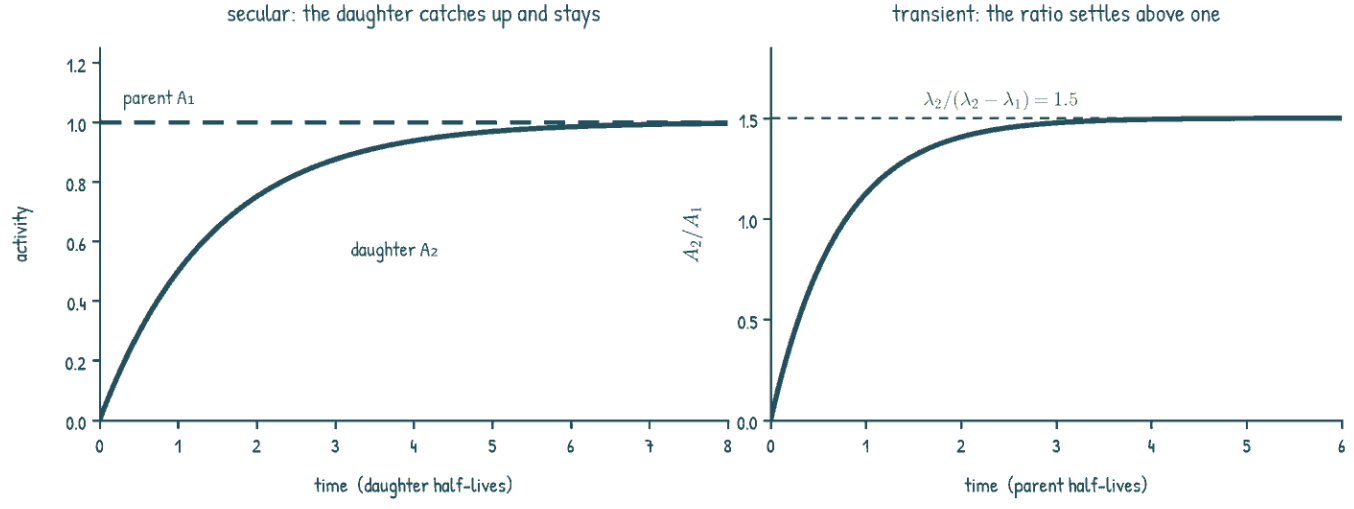
→ elute the daughter at 14 h, not at the end of the day

KEY RESULT
the regimes
Secular: parent ≫ daughter, ratio 1.
Transient: ratio $\lambda_2/(\lambda_2 - \lambda_1)$.
No equilibrium if the parent is the shorter one.

GATE
what the formulas assume
Closed system, no daughter at t = 0, and a single decay path for each nuclide.

TRAP
which is parent
Comparing half-lives backwards. The longer-lived isotope is the one that sets the chain's pace.

Secular equilibrium gives $A_2 = A_1$; transient gives $A_2/A_1 = \lambda_2/(\lambda_2 - \lambda_1)$



KEEP / secular: $A_2 = A_1$; transient: $A_2/A_1 = \lambda_2/(\lambda_2 - \lambda_1)$

Where does a gigawatt come from?

? One fission releases 200 MeV. Multiply by the fission rate and a gram of uranium outruns a tonne of coal.

1 The energy budget of one split

A thermal neutron leaves U-236 excited above its fission barrier; the drop necks and Coulomb wins. The fragments fly apart with most of the energy; the rest arrives later as neutrons, betas and gammas.

$$^{235}\mathrm{U} + n \rightarrow {}^{236}\mathrm{U}^{*} \rightarrow X + Y + 2n + 200\ \mathrm{MeV}$$

→ ≈165 MeV fragments, 5 MeV neutrons, 20 MeV betas and gammas (delayed), 10 MeV neutrinos lost

2 Convert to engineering units

Divide 200 MeV into a gram of U-235: 8.2×10¹⁰ J, about 3.4 tonnes of coal. A gigawatt thermal plant therefore burns about a kilogram a day.

$$\frac{10^{9}}{200 \times 1.602 \times 10^{-13}} = 3.1 \times 10^{19}\ \mathrm{s^{-1}}, \qquad \frac{8.6 \times 10^{13}}{8.2 \times 10^{10}} = 1.05\ \mathrm{kg/day}$$

→ the reactor's daily meal fits in a pocket; the waste does not

3 The chain needs a multiplication factor

k compares neutrons in one generation with the last. Below one the fire dies, at one it is steady, above one it rises. A reactor is designed to sit at k = 1 with delayed neutrons giving the control loop time to act; a bomb is prompt-critical by construction.

$$k < 1 \text{ subcritical}, \qquad k = 1 \text{ steady}, \qquad k > 1 \text{ rising}$$

→ moderator slows, control rods absorb, coolant carries heat, shielding guards the outside

4 A worked power balance

A 1 GW electric plant at 33 % efficiency.

$$P_{\mathrm{th}} = 3.0\ \mathrm{GW}: \quad \frac{3 \times 10^{9}}{200(1.602 \times 10^{-13})} = 9.4 \times 10^{19}\ \mathrm{s^{-1}}$$

$$m = \frac{3(1.05)}{1} = 3.2\ \mathrm{kg/day\ of\ U{-}235}$$

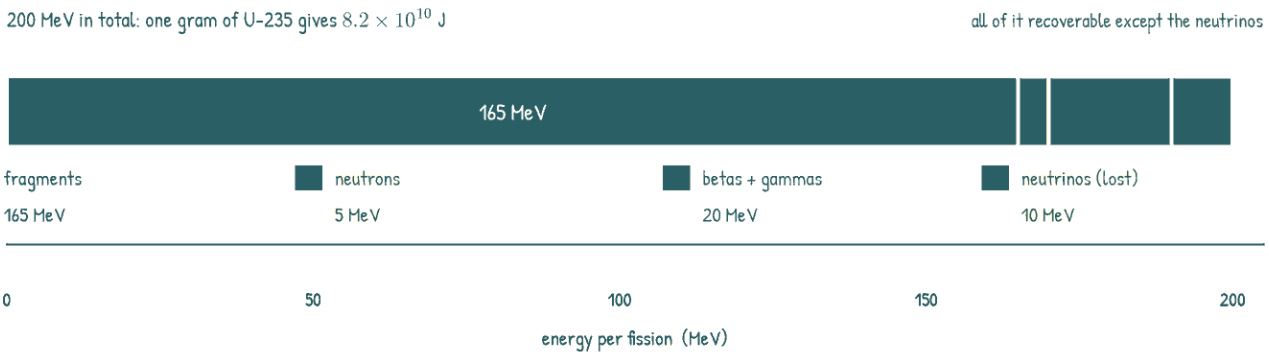
→ three tonnes of coal would do the same job — the factor 10⁴ in one line

KEY RESULT
the arithmetic
200 MeV per fission.
8.2×10¹⁰ J per gram of U-235.
3.1×10¹¹ fissions per second per GW thermal.

GATE
k and the reactor
A power reactor cannot detonate like a weapon: dilute fuel, delayed neutrons, negative feedbacks.
That is not a claim that accidents are impossible.

TRAP
the two units
Confusing thermal and electrical power (factor ≈3), and quoting 200 MeV per *nucleus* as if all the fuel fissioned.

Fragments carry most of it; the neutrinos leave with 10 MeV and are never recovered



KEEP / 200 MeV per fission: 8.2×10¹⁰ J per gram, 3.1×10¹¹ fissions per second per GW

0.1 eV — 1 MeV ▼ — 10 MeV

Why is the Sun not ten billion kelvin?

? Protons repel hard. Classical physics demands an impossible temperature; the Sun runs on the tail of a distribution.

1 The barrier is real and it is large

Two protons at 2 fm sit on a Coulomb hill of 0.72 MeV. Setting k_BT equal to it gives 8×10⁹ K — the classical answer, five hundred times the core.

$$U = \frac{ke^2}{2\ \text{fm}} = 0.72\ \text{MeV} \ \Rightarrow\ T = \frac{U}{k_B} \approx 8 \times 10^9\ \text{K}$$

→ that is the roll-over temperature, not the operating temperature

2 Tunnelling and the Maxwell tail meet in the middle

The reaction rate is the product of how many particles have energy E (a falling exponential) and how likely they are to tunnel through the barrier (a rising one). The product peaks at the Gamow energy, far above k_BT and far below the barrier.

$$E_0 = \left(\frac{E_G (k_B T)^2}{4}\right)^{1/3} \approx 6\ \text{keV} \quad \text{at } 15\ \text{MK}$$

→ the Sun burns at 1.5×10⁷ K because its particles tunnel, not because they climb

3 The chain, and what it pays

The p-p chain fuses four protons in three steps; the first is weak and slow, and it sets the Sun's clock. The 26.7 MeV appears as light and neutrinos.

$$4p \rightarrow {}^4\text{He} + 2e^+ + 2\nu + 26.7\ \text{MeV}, \qquad \dot{m} = \frac{L}{c^2} = 4.3 \times 10^9\ \text{kg/s}$$

→ only 0.7 % of the fused mass becomes energy — and that is still 4 million tonnes a second

4 A worked barrier

Deuterium plus tritium, and the classical temperature of the proton barrier.

$$Q = (0.018883\ \text{u})(931.5) = 17.6\ \text{MeV}$$

$$T = \frac{0.72\ \text{MeV}}{8.617 \times 10^{-11}\ \text{MeV/K}} = 8.4 \times 10^9\ \text{K}$$

→ the D-T reaction is the easiest one, and it still needs 100 MK in the lab

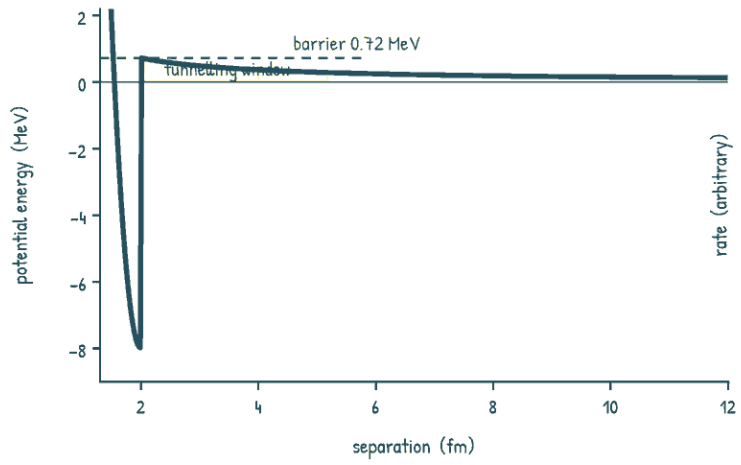
KEY RESULT
the three numbers
Barrier 0.72 MeV at 2 fm.
Core 1.5×10⁷ K, Gamow peak ~6 keV.
26.7 MeV per helium; 4.3×10⁹ kg/s lost.

GATE
when the chain applies
p-p dominates cool stars; the CNO cycle takes over above ~1.7×10⁷ K with carbon as catalyst.
Both give the same net energy.

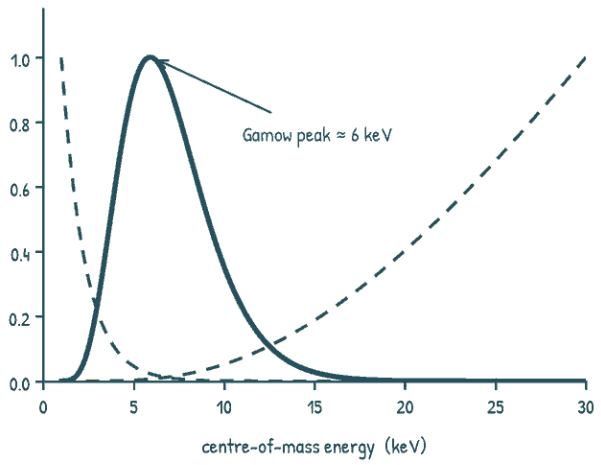
TRAP
the one-step Sun
Four protons do not collide at once. Treating the chain as a single reaction loses the weak first step that sets the lifetime.

Classically the Sun needs 8×10⁹ K; tunnelling and the Maxwell tail make 1.5×10⁷ K enough

the Coulomb hill, and the well behind it



tail × tunnel = the working window



KEEP / the Coulomb barrier needs 8×10⁹ K classically; tunnelling plus the Maxwell tail makes 1.5×10⁷ K enough

0.1 eV ▼50 keV 10 MeV

How does a tube voltage become a shortest wavelength?

? One electron, one photon, and a sharp edge that measures h/e.

1 Making X-rays is thermionic emission plus a cliff

A filament boils electrons off; tens of kilovolts accelerate them into a heavy target; they brake in the nuclear field and radiate. Only about one per cent of the electron power becomes X-rays — the rest heats the anode, which is why the anode rotates.

$$\eta \approx 10^{-9} ZV: \quad \text{W, } 100 \text{ kV} \Rightarrow 0.7\ \%$$

→ a 700 W dental tube radiates a few watts and must dump the other 695 W

2 The cutoff is an energy statement

A braking electron may shed any fraction of its energy, so the spectrum is continuous. But no photon can carry more than the whole electron energy, and that bound is reached when one electron gives everything in one collision.

$$hf_{\max} = eV \Rightarrow \lambda_{\min} = \frac{hc}{eV} = \frac{1.24 \times 10^6}{V\,[\text{V}]}\ \text{pm}$$

→ 30 kV → 41.3 pm; 50 kV → 24.8 pm; 100 kV → 12.4 pm

3 Why the edge is sharp and the tail is soft

The short-wavelength edge is a kinematic extreme: nothing can be shorter, so the distribution starts abruptly there. The long-wavelength side fades because soft photons are reabsorbed in the target and the window before they escape.

→ changing the target changes brightness roughly as Z, never the edge

4 A worked cutoff check

A tube runs at 50 kV. What is the shortest wavelength, and what does 100 kV do to it?

$$\lambda_{\min} = \frac{1.24 \times 10^6}{5 \times 10^4} = 24.8\ \text{pm}$$

$$100\ \text{kV}: \ 12.4\ \text{pm}$$

→ plot λ_min against 1/V and the slope is hc/e: the Duane-Hunt measurement of h

KEY RESULT

the tube in three lines

$$\lambda_{\min} = hc/eV$$

$$\eta \approx 10^{-9} ZV$$

edge from V, height from Z and mA

TRAP

two confusions

Applying λ_min to a characteristic line — it bounds only the continuous spectrum.

Thinking higher Z shifts the edge. It raises the hump.

NUMBERS

tube bench

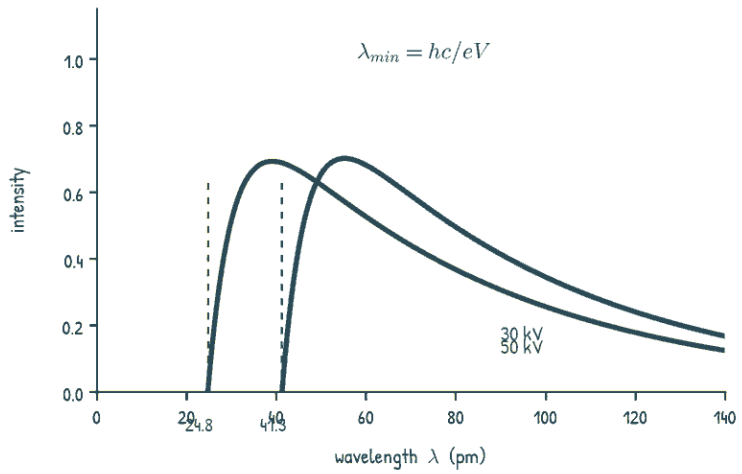
1.24×10⁶ pm·V rule

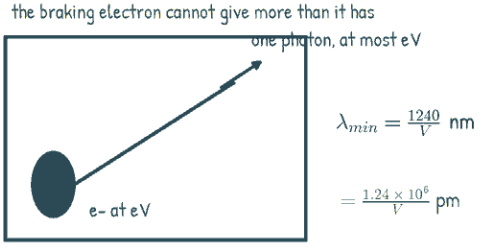
tungsten Z = 74, melting point 3422 °C

1 % efficiency at 100 kV

Duane-Hunt: the shortest X-ray is one electron spending everything at once

voltage sets the edge, target sets the height





target Z changes brightness, never the cutoff

KEEP / λ_min = hc/eV: the edge depends on voltage alone, never on the target



When do the sharp lines appear?

? A spectrum has two parts with two different masters: voltage and the atom.

1 Two mechanisms in one spectrum

Braking radiation gives the continuous hump, cut off at hc/eV. Separately a fast electron can eject an inner one; the vacancy refills from above and the photon carries a level difference — a sharp, element-specific line.

$$E_{K\alpha} = E_L - E_K, \qquad E_{K\beta} = E_M - E_K$$

→ the lines sit where the atom says; the hump sits where the voltage says

2 The threshold logic examiners test

A K line needs a K vacancy, and making one costs at least the K binding energy. Below that voltage the lines are absent however large the current; above it they appear and grow.

$$V_{\text{th}} = \frac{E_K}{e}: \quad \text{Cu } 9\text{ kV},\ \text{Mo } 20\text{ kV},\ \text{W } 70\text{ kV}$$

→ no lines means below threshold, not a failed tube

3 Screened hydrogen, and Moseley

The falling L electron sees a nucleus of charge Z screened by the one remaining K electron, so the transition is hydrogenic arithmetic with (Z − 1) for Z. The square root linearises it: a straight line in Z — how Moseley ordered the elements.

$$E_{K\alpha} = \tfrac{3}{4}(13.6)(Z-1)^2 \text{ eV}, \qquad \sqrt{f} = \sqrt{\tfrac{3}{4}cR}\,(Z-1)$$

→ copper: 8.0 keV predicted, 155 pm against a measured 154 pm

4 A worked threshold

Copper's K shell binds 8.98 keV. What is the minimum tube voltage for its Kα line?

$$V_{\text{th}} = \frac{8.98\text{ keV}}{e} = 8.98\text{ kV}$$

$$V = 4.5\text{ kV}: \ \text{cutoff } \frac{1.24\times10^6}{4500} = 276\text{ pm},\ \text{no lines}$$

→ below threshold the continuous spectrum survives and the lines vanish: two masters, two behaviours

KEY RESULT
line and threshold

$$E_{K\alpha} = \tfrac{3}{4}(13.6)(Z-1)^2$$

$$\sqrt{f} \propto (Z-1)$$

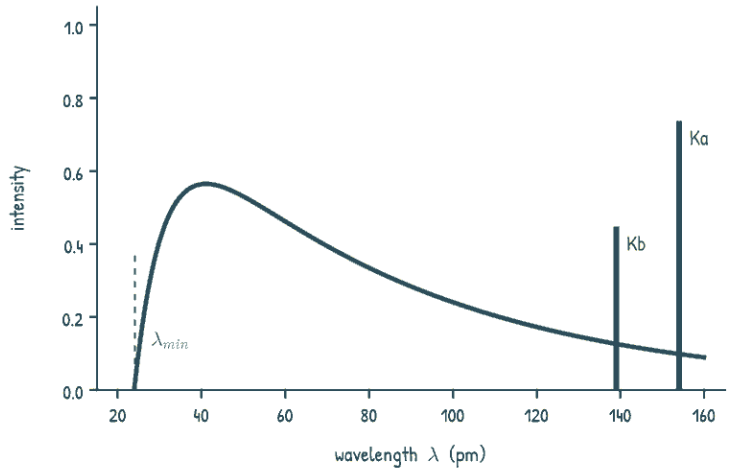
V_th = E_K/e; below it, no lines

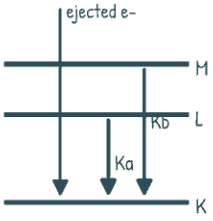
GATE
what the shielded form assumes
One K electron left, so it works for K lines of heavy atoms.
L lines need b ≈ 7.4, so the constant is line-specific.

TRAP
the two masters
Raising current brightens both parts; raising voltage moves the edge and can switch lines on. Neither moves a line's wavelength.

Characteristic lines are the atom fingerprint; Moseley turned them into a periodic table ruler

lines ride on the bremsstrahlung hump





$E_{K\alpha} = \tfrac{3}{4}(13.6)(Z-1)^2$ eV

$\sqrt{f} \propto (Z-1)$

Cu: K binds 9 keV, Kα = 8.0 keV

the vacancy must be affordable: V > 9 kV

KEEP / the hump is bremsstrahlung; the spikes need a vacancy, so they need a threshold voltage

0.1 eV — 100 keV — 10 MeV

Why does a radiograph show bone and not skin?

? Attenuation is exponential, and the exponent is roughly Z cubed.

1

A fixed fraction per layer

Every additional slice of material removes the same fraction of what is left, because it presents the same number of absorbers. That sentence is the differential equation, and its solution is the exponential.

$$I = I_0 e^{-\mu x}, \qquad x_{1/2} = \frac{\ln 2}{\mu}$$

→ lead at 100 keV: $x_{½}$ = 0.12 mm, so μ = 58 cm⁻¹

2

Why bone casts a shadow

Photoelectric absorption scales roughly as Z^3–Z^4 and falls with photon energy. Calcium (Z = 20) against carbon and oxygen (Z = 6, 8) is therefore a contrast of order twenty in μ, which the film records as a shadow. Barium and iodine meals exploit the same appetite.

→ soft X-rays give better contrast and more dose: one curve read two ways

3

The engineering trade

The same μ that creates contrast also deposits energy, and the energy deposited per unit mass is the dose. Thicker patients need harder beams, harder beams give less contrast, and the compromise is what a radiology protocol is. Computed tomography is this law inverted numerically over many projections.

$$\text{dose} \propto \int \mu \, \Phi \, dt$$

4

A worked half-value stack

Lead at 100 keV has $x_{½}$ = 0.123 mm. What thickness leaves one per cent?

$$n = \frac{\log(0.01)}{\log(0.5)} = 6.64 \text{ half layers}$$

$$x = 6.64(0.123) = 0.82 \text{ mm}$$

→ a lead apron is 0.25–0.5 mm: enough for scattered diagnostic radiation, not for a beam

KEY RESULT

the law

$$I = I_0 e^{-\mu x}$$

$$x_{1/2} = \ln 2 / \mu$$

μ rises as Z^3–Z^4, falls with E

GATE

narrow beam only

μ is measured with a narrow beam and a narrow detector. A broad beam collects scattered photons and needs a build-up factor.

NUMBERS

half-value layers

Pb 0.12 mm at 100 keV

Al a few mm at 50 keV

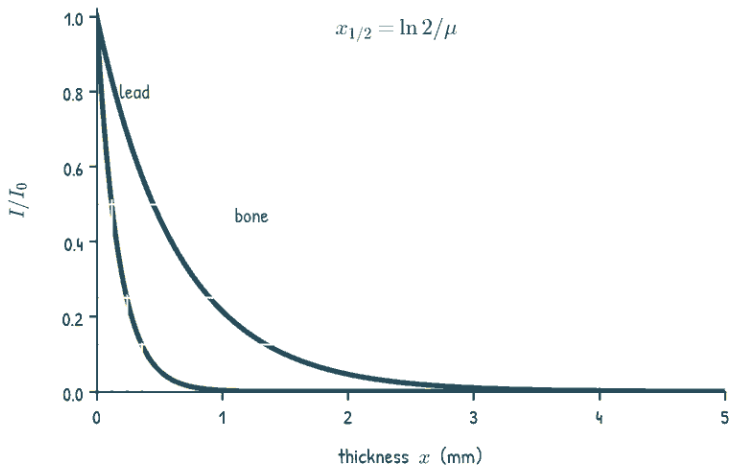
tissue ~ cm at 50 keV

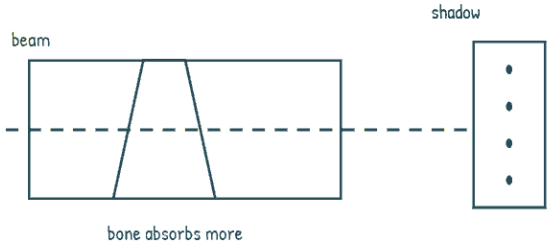
Exponential attenuation: the radiograph is a map of mu times thickness

every layer removes the same fraction

$x_{1/2} = \ln 2 / \mu$

$\mu \propto Z^3 - Z^4$, falls with E





bone absorbs more

contrast and dose are the same curve, read two ways

KEEP / I = I₀ e^{−μx}: contrast comes from μ, and μ is a steep function of Z and E

0.1 eV ▼8 keV 10 MeV

How do you measure an atom with a wavelength?

? Bragg's law turns angles into spacings — and spacings into Avogadro's number.

1 Only a crystal is fine enough

A grating resolves structure of order its own spacing. Visible light (500 nm) sees atomic planes (0.3 nm) as a smooth surface — the ratio is 10^3 too coarse for any Bragg angle. X-rays at 100 pm match the spacing, so the crystal becomes the grating.

→ Laue's 1912 spots proved both that X-rays are waves and that crystals are lattices

2 The path difference, with the angle made foolproof

Two rays strike successive planes at glancing angle θ. The lower one travels an extra $d\sin\theta$ going in and the same coming out. Constructive interference needs that total to be a whole number of wavelengths.

$$2d\sin\theta = n\lambda, \qquad n \le \frac{2d}{\lambda}$$

→ θ is measured from the plane; the detector sits at 2θ from the beam

3 From a spacing to Avogadro

In rock salt the cubic cell alternates ions, so each ion pair owns a volume $2d^3$ where d is the neighbouring-ion spacing. A balance gives the same volume as $M/\rho N_A$, and equating them yields Avogadro's number from a wavelength and an angle.

$$N_A = \frac{M}{2\rho d^3} = \frac{0.05844}{2(2165)(2.82\times10^{-10})^3}$$

$$= 6.02\times10^{23}\ \text{mol}^{-1}$$

→ the missing reflections are structure-factor information, not experimental error

4 A worked Avogadro

NaCl: d = 0.282 nm, ρ = 2165 kg/m³, M = 58.44 g/mol.

$$N_A = \frac{0.05844}{2(2165)(2.82\times10^{-10})^3} = \frac{0.05844}{9.71\times10^{-26}}$$

$$= 6.02\times10^{23}\ \text{mol}^{-1}$$

→ a wavelength, an angle and a balance reproduce a constant measured in electrochemistry

KEY RESULT

the law and its use

$$2d\sin\theta = n\lambda$$

$$d = \frac{n\lambda}{2\sin\theta}$$

$$N_A = M/2\rho d^3$$

TRAP

the convention

θ from the plane, so the path difference is 2d sinθ. Writing 2d cosθ is the single most common error in this topic.

And the detector angle is 2θ.

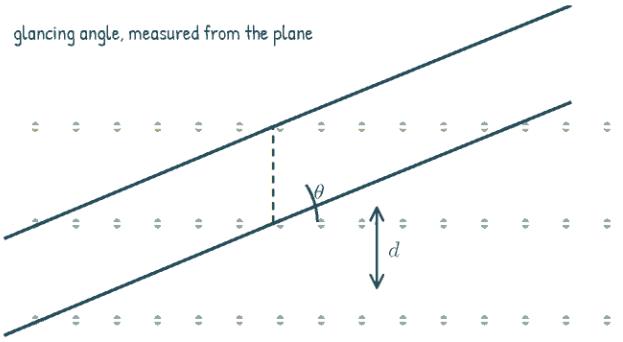
NUMBERS

NaCl bench

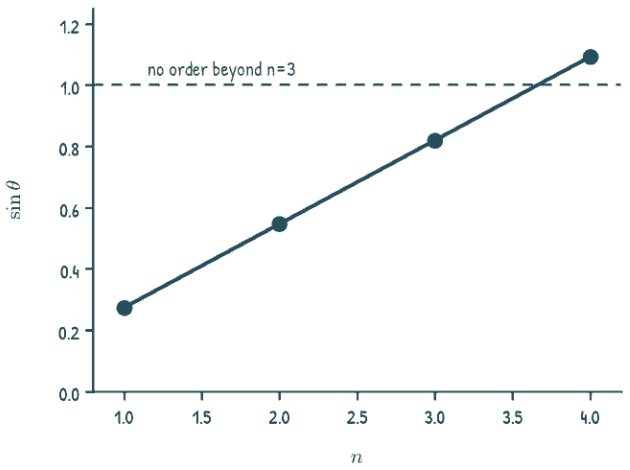
d = 282 pm, a = 564 pm

ρ = 2165 kg/m³, M = 58.44 g/mol

Bragg: path difference 2d sin(theta); orders stop when sin(theta) would exceed one



$2d\sin\theta = n\lambda$



KEEP / 2d sinθ = nλ with θ from the plane: an angle, a wavelength, and a lattice constant

What happens when a photon hits an electron?

? Billiards with light: the shift is absolute, angle-only, and independent of everything else.

1 The experiment that classical physics cannot survive

Compton scattered 100 pm X-rays off graphite and found two peaks at every angle: one at the original wavelength, one shifted by an amount that grows with angle and depends on nothing else. Classical electrodynamics predicts a single unshifted line at every angle.

→ the unshifted peak is whole-atom recoil off tightly bound electrons — not an error

2 Conservation closes the triangle

Write energy and both momentum components, eliminate the electron angle, and use the relativistic energy-momentum relation. What falls out is a shift that depends only on the scattering angle and one constant with the dimensions of length.

$$\Delta\lambda = \frac{h}{m_e c}(1-\cos\theta) = 2.426(1-\cos\theta)\ \text{pm}$$

→ 0° → 0; 90° → 2.43 pm; 180° → 4.86 pm, the maximum

3 The electron takes the rest

Whatever the photon loses, the electron gains as kinetic energy. Writing x = hf/m_ec², the recoil energy has a clean closed form, and the electron always goes forward of the perpendicular — it can never come back.

$$K_e = hf\frac{x(1-\cos\theta)}{1+x(1-\cos\theta)}, \qquad \tan\phi = \frac{1}{(1+x)\tan(\theta/2)}$$

4 A worked back-scatter

A 100 keV photon is scattered through 180°.

$$\Delta\lambda = 4.86\ \text{pm}, \qquad x = \frac{100}{511} = 0.196$$

$$K_e = 100\frac{2(0.196)}{1+2(0.196)} = 28\ \text{keV}$$

→ the maximum energy transfer is at 180°, and it is still well below the photon energy

KEY RESULT

the shift

$$\Delta\lambda = \frac{h}{m_e c}(1-\cos\theta)$$

λ_C = 2.426 pm, a fundamental length

Absolute shift; the fraction varies with λ.

TRAP

three misreadings

Treating Δλ as a percentage of λ.

Explaining the unshifted peak as failure of the model.

Using p = E/c for the recoil electron.

NUMBERS

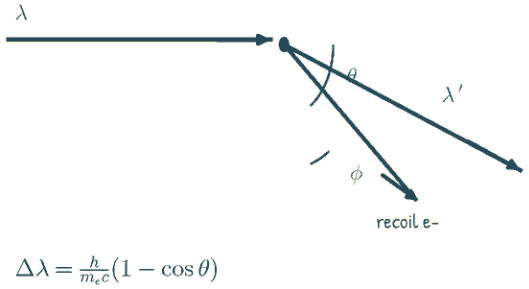
the scale

visible light: shift 10^{-5} of λ — invisible

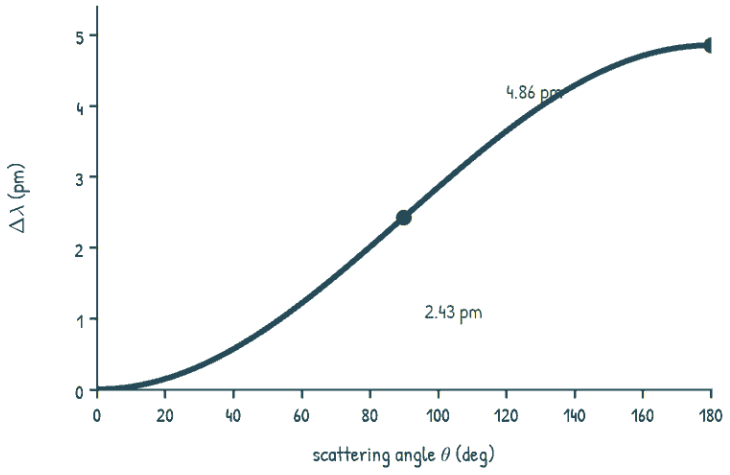
100 keV: shift is 2 % of λ

1 MeV: shift is 30 % of λ

Compton: the shift depends on angle alone - absolute, never a percentage



$$\Delta\lambda = \frac{h}{m_e c}(1-\cos\theta)$$



KEEP / $\Delta\lambda = 2.426(1-\cos\theta)$ pm: an absolute shift, never a percentage

0.1 eV 100 keV 10 MeV

Can the Compton shift be derived without trigonometry?

? Yes — square the four-momentum of the electron and the algebra collapses.

1 Four-vectors make conservation linear

Energy and the three momentum components are one object. Conservation of four-momentum is then a single vector equation: the incoming photon equals the outgoing photon plus the recoil electron.

$$p_\gamma = p'_\gamma + p_e \;\Rightarrow\; p_e = p_\gamma - p'_\gamma$$

2 Square it and use the mass shell

The electron's four-momentum squared is $m_e^2c^2$, a number every frame agrees on. The photon's is zero. Expanding the square leaves only the dot product of the two photon momenta, which in the lab frame is a product of energies and a cosine.

$$(p_\gamma - p'_\gamma)^2 = m_e^2c^2 \;\Rightarrow\; 2p_\gamma \cdot p'_\gamma = 2m_e^2c^2$$

$$\frac{1}{\lambda'} - \frac{1}{\lambda} = \frac{1-\cos\theta}{\lambda_C}$$

→ two lines of algebra, no recoil angle, no trigonometric identities

3 Why this route is worth learning

The scalar derivation needs three conservation equations and one elimination. The invariant route needs one equation. When a problem has several bodies and several angles, squaring a four-momentum difference is almost always the shortest path — and it generalises to pair production and to any two-body decay.

→ same answer, but the method scales where the trigonometry does not

4 A worked invariant

Use the same invariant on a 2 MeV photon scattering at 60°.

$$\Delta\lambda = 2.426(1-\cos 60^\circ) = 1.21 \text{ pm}$$

$$\lambda = 0.62 \text{ pm} \;\Rightarrow\; \lambda' = 1.83 \text{ pm}$$

→ the fractional shift is 195 %: at MeV energies the photon loses most of its energy

KEY RESULT

the invariant

$$p_e = p_\gamma - p'_\gamma$$

$$p^2 = m^2c^2$$

Square first, ask questions later.

INSIGHT

why thresholds are higher

In a fixed-target collision the products must carry the target's momentum, so the centre-of-mass energy is √s, not the beam energy plus the target mass.

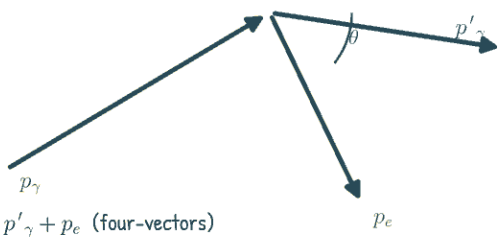
TRAP

mixing frames

The dot product p·p′ must be evaluated in one frame. Using the lab energies but a centre-of-mass angle is the standard error.

One invariant replaces three conservation equations: the Compton shift falls out in two lines

close the triangle with four-momenta



$p_\gamma = p'_\gamma + p_e$ (four-vectors)

the invariant route

square the electron line:

$$(p_\gamma - p'_\gamma)^2 = m_e^2c^2$$

$$2p \cdot p' = 2m_e^2c^2$$

$$\Rightarrow\; \Delta\lambda = \frac{h}{m_ec}(1-\cos\theta)$$

two lines, no trigonometry

KEEP / one invariant replaces three conservation equations and two eliminands

0.1 eV ▼1 keV 10 MeV

How does a fast electron brighten a dull photon?

? The Compton event run backwards, and the reason radio galaxies shine in X-rays.

1 Run the collision backwards

In Compton scattering a photon hands energy to an electron. If the electron is already moving fast and the photon meets it head-on, the photon gains energy instead. In the electron's frame this is ordinary Compton scattering; transform back and the photon is boosted.

$$\epsilon' \approx \gamma^2 \epsilon \quad (\gamma\epsilon \ll m_e c^2)$$

→ a radio photon at Γ = 1000 becomes a keV photon: radio galaxies glow in X-rays

2 The moving-mirror picture

Treat the electron as a mirror approaching the light. One Doppler shift takes the photon into the electron frame, a second takes it back out. Two shifts of order Γ give the square.

$$\frac{\epsilon'}{\epsilon} = \frac{1+\beta}{1-\beta} \approx 4\gamma^2 \quad (\beta \to 1)$$

→ the same factor 4Γ² that appears in synchrotron and in the Doppler beaming of jets

3 Where it matters, and where it stops

Inverse Compton sets the X-ray output of jets, of the cosmic microwave background seen by hot cluster gas, and of any source with relativistic electrons and a soft photon bath. Once Γε approaches m_ec² the process enters the Klein-Nishina regime and the cross-section falls — that is where the simple boost stops.

→ the Sunyaev-Zel'dovich effect is this process on the microwave background

4 A worked boost

Microwave photons at 6×10^{-4} eV hit electrons with Γ = 10^3.

$$\epsilon' \approx 4\gamma^2\epsilon = 4(10^6)(6\times10^{-4}) = 2.4\times10^3 \text{ eV}$$

$$2.4 \text{ keV} = \text{ soft X ray}$$

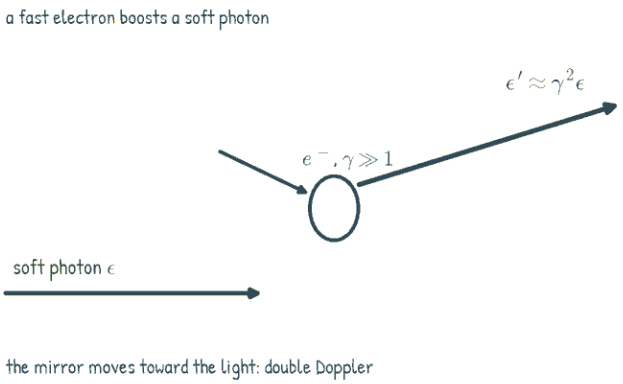
→ the same electrons synchrotron-radiate at keV too: two channels, one population

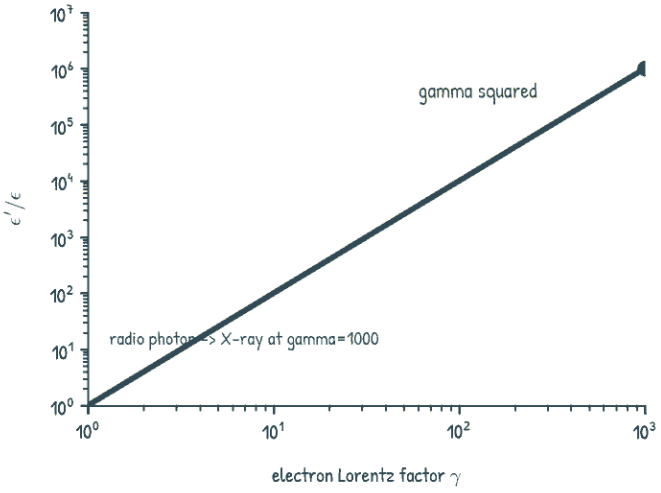
KEY RESULT
the boost
$$\epsilon' \approx \gamma^2\epsilon$$
$$\propto 4\gamma^2 \text{ at } \beta \to 1$$
two Doppler shifts, not one

GATE
when it fails
Γε ≳ m_ec²: recoil matters, the cross-section drops (Klein-Nishina).
Also assumes an isotropic soft photon bath in the electron frame.

TRAP
one shift only
Using 2Γ instead of 4Γ². The photon is boosted twice: in and out.

Inverse Compton: the same kinematics run backwards, and astrophysics uses it





KEEP / a photon bouncing off a relativistic electron gains of order Γ² in energy

When does a photon stop being a photon?

? Above 1.022 MeV it can become matter — but never in empty space.

1

The threshold is a rest-mass statement

Creating an electron-positron pair costs $2m_ec^2$ = 1.022 MeV of rest energy before any kinetic energy. A photon below that energy cannot do it, whatever the material.

$$E_{\min} = 2m_ec^2 = 1.022\ \text{MeV}, \qquad K_{\text{pair}} = E_\gamma - 1.022\ \text{MeV}$$

→ a 2 MeV photon leaves 0.98 MeV to share between the two particles

2

Why empty space is forbidden

In free space, energy and momentum cannot both be conserved by one photon splitting into two equal-mass particles: in the centre-of-mass frame the photon has momentum but the pair would have none. A nucleus (or an electron) must take the recoil.

$$p_\gamma \neq p_+ + p_- \ \text{ in any frame with } E_\gamma = 2m_ec^2$$

→ the same reason a single photon cannot produce a single electron

3

The three experiments that complete the photon

Photoelectric effect proves the energy comes in quanta. Diffraction proves the wave. Compton proves the momentum. Pair production closes the circle: at high enough energy the quantum can be converted into the very particles whose mass it never had.

→ in a bubble chamber a photon leaves no track until it converts — then two curly ones appear

4

A worked threshold

A 2 MeV photon converts near a nucleus.

$$K_{\text{total}} = 2.00 - 1.022 = 0.98\ \text{MeV}$$

$$\text{shared as } K_+ + K_- = 0.98\ \text{MeV}$$

→ in a magnetic field the two tracks curl opposite ways with the same radius if the sharing is equal

KEY RESULT

the threshold

$$2m_ec^2 = 1.022\ \text{MeV}$$

$$K = E_\gamma - 1.022$$

needs a third body for momentum

TRAP

free-space conversion

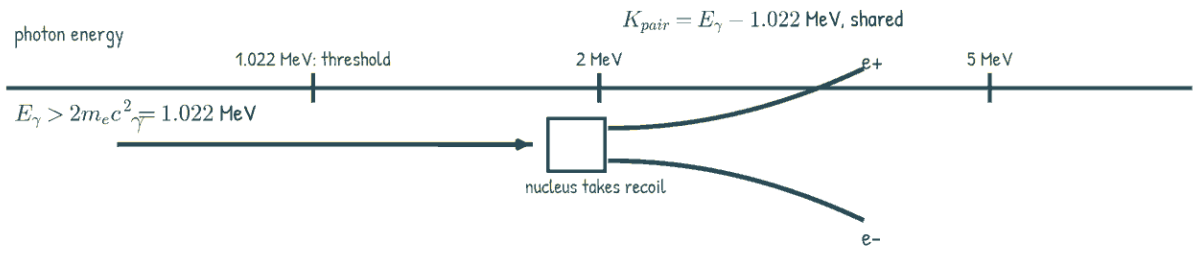
Allowing $\gamma \rightarrow e^+e^-$ with nothing else present. Momentum conservation forbids it in every frame.

INSIGHT

the photon, completed

Energy quantum (photoelectric), wave (diffraction), momentum (Compton), convertibility (pair production). Four experiments, one object.

Pair production needs 1.022 MeV and a nucleus to carry the momentum



KEEP / E > $2m_ec^2$ and a third body: energy alone is not enough, momentum must balance

How did a straight line predict missing elements?

? Moseley plotted √f against Z in 1913 and the periodic table became a measurement.

1 Why the line is straight

Kα is a 2 → 1 transition in a nucleus screened by the one remaining K electron. The energy carries $(Z-1)^2$, so the frequency carries it too, and its square root is linear in Z.

$$f = \tfrac{3}{4}cR(Z-1)^2 \;\Rightarrow\; \sqrt{f} = \sqrt{\tfrac{3}{4}cR}\,(Z-1)$$

→ slope √(3cR/4) = 4.97×10⁷ (Hz)^{1/2} per unit Z

2 What the intercept tells you

The straight line does not pass through the origin: it hits √f = 0 at Z = 1, not Z = 0. That offset is the shielding constant, and the same trick with L lines gives b ≈ 7.4 because more electrons screen.

$$b_{K\alpha} \approx 1, \qquad b_{L\alpha} \approx 7.4$$

→ Moseley's "a and b" are exactly the slope and the shield

3 The prediction that found four elements

Because the line is monotonic and every element must sit on it, a gap in the sequence is a missing element rather than a gap in nature. Elements 43, 61, 72 and 75 were identified this way before they were isolated — technetium, promethium, hafnium, rhenium.

→ it also settled the ordering: Z, not atomic weight, is the periodic variable

4 A worked prediction

Predict the Kα wavelength of silver (Z = 47) from copper's 154 pm.

$$\lambda \propto \frac{1}{(Z-1)^2}: \;\; \lambda_{Ag} = 154\frac{28^2}{46^2} = 57 \text{ pm}$$

$$f = \tfrac{3}{4}cR(46)^2 = 5.3\times10^{18} \text{ Hz}$$

→ the measured value is 56 pm: one line, one constant, every element

KEY RESULT

Moseley

$$\sqrt{f} = a(Z-b)$$

Kα: b ≈ 1; Lα: b ≈ 7.4

Z is the ordering variable, not mass.

INSIGHT

what it changed

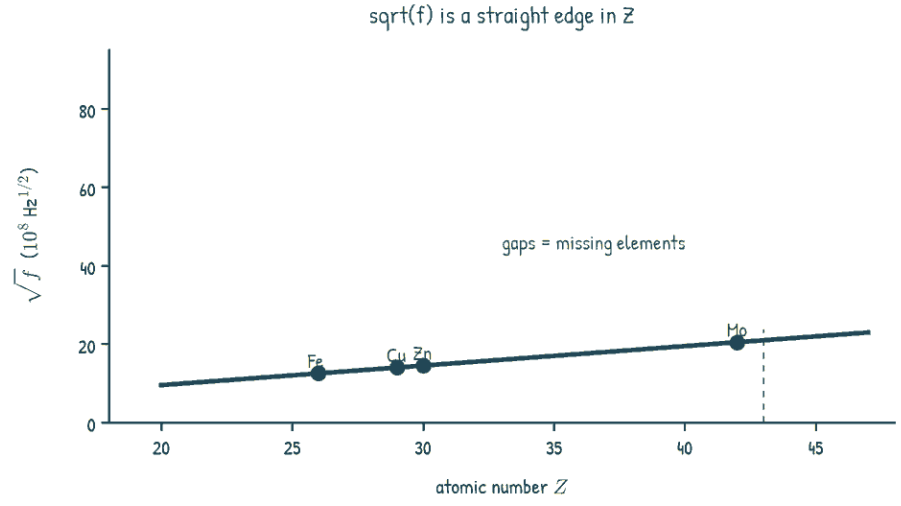
Before 1913 the periodic table was a chemical pattern with guessed gaps. After it, the gaps were places with a known X-ray address.

TRAP

one line, one constant

Using a Kα value of b for an L line, or treating b as universal.

Moseley: the K-alpha frequency is a ruler for Z, with the missing elements as gaps



$$E_{K\alpha} = \tfrac{3}{4}(13.6)(Z-1)^2$$

one K electron

screens the nucleus

L lines use

$b \approx 7.4$

so the constant is line-specific

KEEP / √f ∝ (Z − b): every element is a point on one line, and gaps mean missing elements

▼13.6 eV
0.1 eV 10 MeV

How long would a classical atom live?

? Sixteen picoseconds, and a continuous spectrum while it dies. That is the crisis.

1 An accelerating charge radiates

Larmor's formula gives the power radiated by an accelerating charge. For an electron in a circular orbit the acceleration is v^2/r, so the power climbs steeply as the orbit shrinks — which makes the collapse accelerate as it proceeds.

$$P = \frac{e^2 a^2}{6\pi\varepsilon_0 c^3}, \qquad a = \frac{ke^2}{m_e r^2}$$

2 Integrate the energy loss

The orbital energy is $-ke^2/2r$, so losing energy means losing radius. Substituting gives a separable equation whose integral is a lifetime proportional to the cube of the starting radius.

$$\frac{dE}{dt} = -P \;\Rightarrow\; r^2\dot{r} = -\frac{2C}{ke^2} \;\Rightarrow\; \tau = \frac{ke^2 r_0^3}{6C}$$

$$\tau(a_0) = 1.6 \times 10^{-11}\ \text{s}$$

→ $\tau_C \approx 5$ mm: a slow spiral of ten million orbits, not a plunge

3 Two predictions that both fail

The atom should live 16 ps, and while dying its orbital frequency should sweep continuously upward, emitting a continuum from infrared to X-rays. Real atoms live forever and emit sharp lines. Both failures are fixed by the same assumption: stationary states.

→ the stability and the line spectrum are one problem, not two

4 A worked collapse time

Evaluate the Larmor-integral lifetime from a_0:

$$C = \frac{e^2}{6\pi\varepsilon_0 c^3}\left(\frac{ke^2}{m_e}\right)^2 = 6.3 \times 10^{-55}\ \text{J m}^4$$

$$\tau = \frac{ke^2 a_0^3}{6C} = 1.6 \times 10^{-11}\ \text{s}$$

→ $\tau_C \approx 5$ mm and about 10^9 orbits: slow, and fatal

KEY RESULT
the collapse

$$P = e^2 a^2 / 6\pi\varepsilon_0 c^3$$

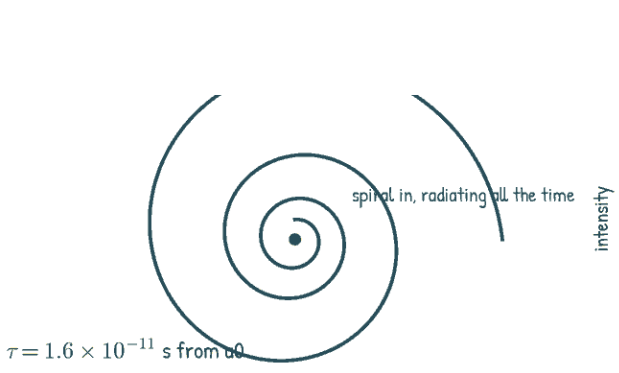
$$\tau = ke^2 r_0^3 / 6C$$

16 ps from a_0; a continuum, not lines

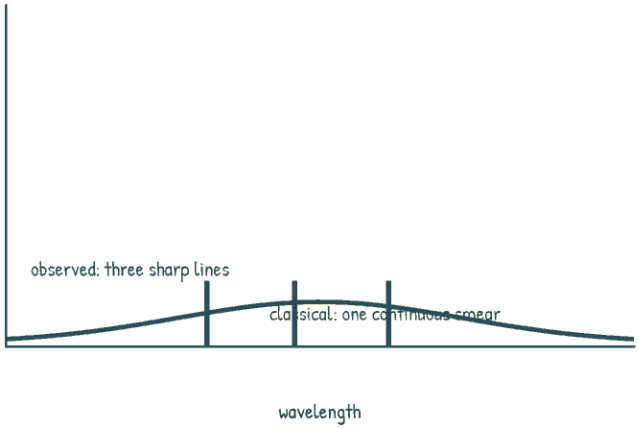
INSIGHT
why it is a spiral
$\tau_C \approx 5$ mm is five orders larger than the atom, so the electron completes ~10^9 orbits while dying. The "instant collapse" story is wrong in scale.

TRAP
the wrong lifetime
Quoting 10^{-8} s — that is an excited-state lifetime, not the classical collapse. The collapse is 10^{-11} s.

The classical atom lives 16 ps and glows a continuum; real atoms do neither



a dying atom cannot be line-bright



KEEP / $\tau \propto r^3$: the collapse is a slow spiral, not a plunge, and it radiates every frequency



Why does quantum mechanics become classical for large n?

? Expand the level spacing and the photon frequency turns into the orbital frequency.

1

A large orbit is a classical antenna

An electron on a big orbit radiates at its orbital frequency. Quantum mechanics says it radiates at the energy gap divided by h. For the two to agree, the gap must approach the orbital energy spacing — which is what a Taylor expansion shows.

$$\Delta E = \frac{A}{n^2}\left[(1-1/n)^{-2}-1\right] \approx \frac{2A}{n^3}\left(1+\frac{3}{2n}\right)$$

$$f_{\text{photon}} = f_{\text{orbit}}\left(1+\frac{3}{2n}\right)$$

2

Check the numbers at n = 100

The correction is 1.5 %. At n = 10 it is 15 %, and the quantum world is visibly non-classical. The ratio approaches one from above, monotonically, and the limit is the statement that quantum mechanics contains classical mechanics.

$$n=100:\ \ f_{\text{orb}} = 6.58\times10^{9}\ \text{Hz},\ \ f_{\text{ph}} = 6.68\times10^{9}\ \text{Hz}$$

→ Rydberg atoms at n ~ 100 are used as sensors precisely because they are almost classical

3

Why this is more than a curiosity

Correspondence is the consistency condition any new theory must satisfy: reduce to the old one where it works. Bohr used it to guess selection rules before the full theory existed.

4

A worked n = 100 check

Compare orbital and photon frequency at n = 100:

$$f_{\text{orb}} = \frac{v}{2\pi r} = 6.58\times10^{9}\ \text{Hz}$$

$$f_{\text{ph}} = \frac{\Delta E}{h} = 6.68\times10^{9}\ \text{Hz}$$

→ ratio 1.015, exactly 1 + 3/(2n) with n = 100

KEY RESULT

the limit

$$\Delta E/h = f_{\text{orb}}(1+3/2n)$$

Correction falls as 1/n

Consistency condition, not a coincidence

GATE

what it does not say

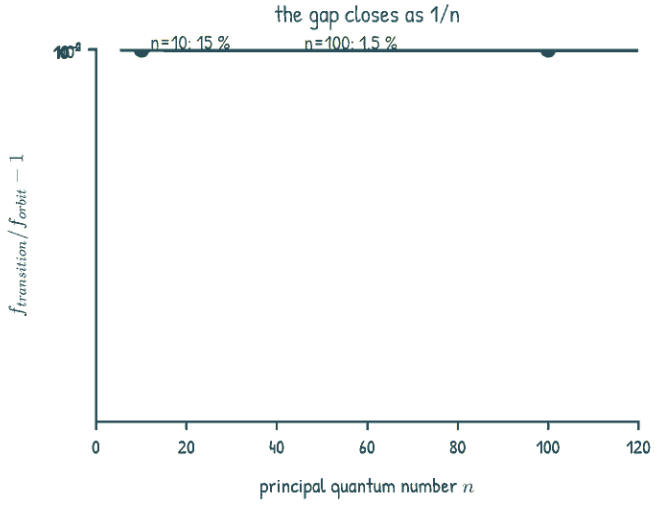
It does not say the classical trajectory is recovered for every state — only for coherent, large-n ones.

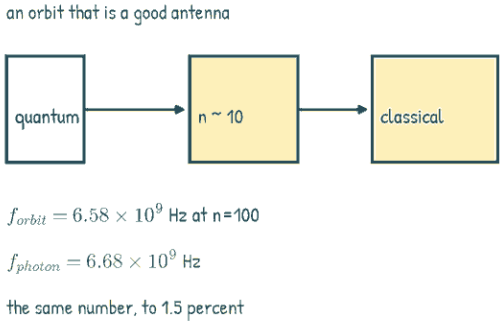
NUMBERS

bench

n = 10: 15 % · n = 100: 1.5 % · n = 150: 1 %

Correspondence: for large n the photon frequency melts into the orbital frequency





KEEP / for large n the gap closes as 1/n, and the photon frequency equals the orbit frequency to 1.5 %

What if the electron weighed 207 times more?

? Muonic hydrogen is a nuclear ruler; positronium is a hydrogen twice as big.

1

The substitution is the whole physics

The two-body problem reduces to one body of reduced mass μ orbiting a fixed centre. Every result of the Bohr derivation then goes through unchanged with m_e replaced by μ, so the radius scales as 1/μ and the binding as μ.

$$\mu = \frac{m_1 m_2}{m_1 + m_2}, \qquad a \propto \frac{1}{\mu}, \qquad E \propto \mu$$

2

Muonic hydrogen: an atom the size of a nucleus

A muon is 207 times the electron mass, so the ground orbit is 256 fm — inside the nuclear charge distribution of a proton. The levels therefore shift by a measurable fraction, and comparing them with the point-nucleus prediction measures the proton radius.

$$a_\mu = \frac{0.529\ \text{Å}}{207} = 256\ \text{fm}, \qquad E_1 = -2.81\ \text{keV}$$

→ this is the "proton radius puzzle": muonic and electronic measurements disagreed for a decade

3

Positronium: half the mass, twice the size

Two equal masses give μ = m/2, so the "atom" is twice as big and half as bound, and every wavelength doubles. It also annihilates — the ground state lives 125 ps — which makes it a laboratory for QED rather than for chemistry.

$$a = 1.06\ \text{Å}, \qquad E_1 = -6.8\ \text{eV}, \qquad \lambda_{2\to1} = 243\ \text{nm}$$

4

A worked positronium line

The 2 → 1 line of positronium:

$$\Delta E = 6.8\left(\tfrac{3}{4}\right) = 5.1\ \text{eV}$$

$$\lambda = \frac{1240}{5.1} = 243\ \text{nm}$$

→ exactly twice hydrogen's 121.6 nm, because μ is exactly half

KEY RESULT
exotic scaling

$$\mu = m_1 m_2/(m_1 + m_2)$$

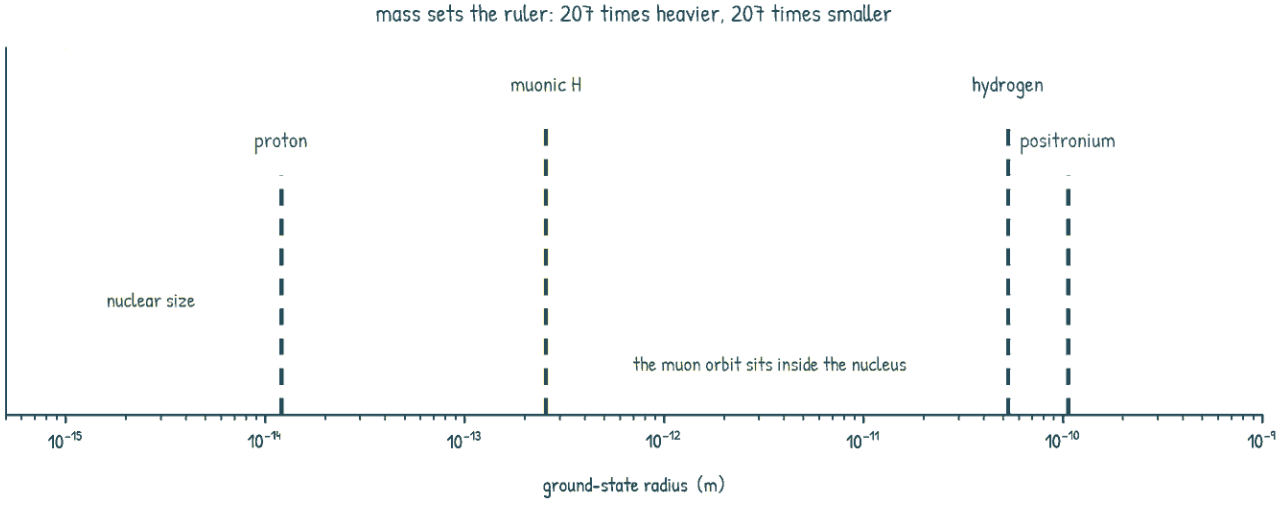
$$a \propto 1/\mu, \quad E \propto \mu$$

E·a is invariant.

INSIGHT
why muonic atoms measure radii
The wavefunction overlaps the nucleus, so the level depends on the charge distribution rather than on a point charge.

TRAP
using m_e
Computing muonic hydrogen with the electron mass gives an answer wrong by 207 — and the proton-radius claim wrong by 207 too.

Exotic atoms are scaling tests: replace the electron mass by the reduced mass



KEEP / every mass-dependent result takes μ: 207 times heavier, 207 times smaller

Can you see energy levels without looking at light?

? Yes: fire electrons at atoms and watch the current develop teeth.

1

The experiment in one sentence

Electrons accelerate through mercury vapour toward a grid. If the vapour had no internal structure they would gain energy continuously and the current would rise smoothly. Instead it dips periodically: at certain energies an electron can hand 4.9 eV to an atom and keep only the remainder, so it no longer has enough to reach the collector.

→ elastic collisions dominate below 4.9 eV; inelastic ones switch on exactly at it

2

Why the dips repeat

After an inelastic collision the electron restarts from near rest. It therefore needs another 4.9 eV of free path before it can excite again, so the collector current dips at every multiple of the threshold voltage.

$$V_{\text{dip}} = n\,(4.9\ \text{V}), \qquad n = 1, 2, 3$$

→ the spacing, not the position, is the measurement: it is insensitive to contact potentials

3

What it proves that spectroscopy does not

Optical spectroscopy sees the photons that come out; Franck-Hertz sees the energy that goes in. Together they close the book: the levels are real, they are discrete, and they are the same levels whether you excite them with electrons or with light. It also shows that excitation costs a quantum, not a continuous trickle.

→ the mercury atom then emits 254 nm ultraviolet as it de-excites — the same 4.9 eV

4

A worked dip position

Where do the first three dips sit, and why not at exactly 4.9 V?

$$V_{\text{dip}} \approx V_{\text{contact}} + n(4.9\ \text{V}), \quad n = 1, 2, 3$$

$$4.9,\ 9.8,\ 14.7\ \text{V plus a constant offset}$$

→ the spacing is the measurement; the offset is contact potential and is subtracted, not explained

KEY RESULT

the measurement

$$\Delta V = 4.9\ \text{V}\ (\text{Hg})$$

dips at $n\Delta V$

inelastic threshold = level spacing

GATE

why the vapour pressure matters

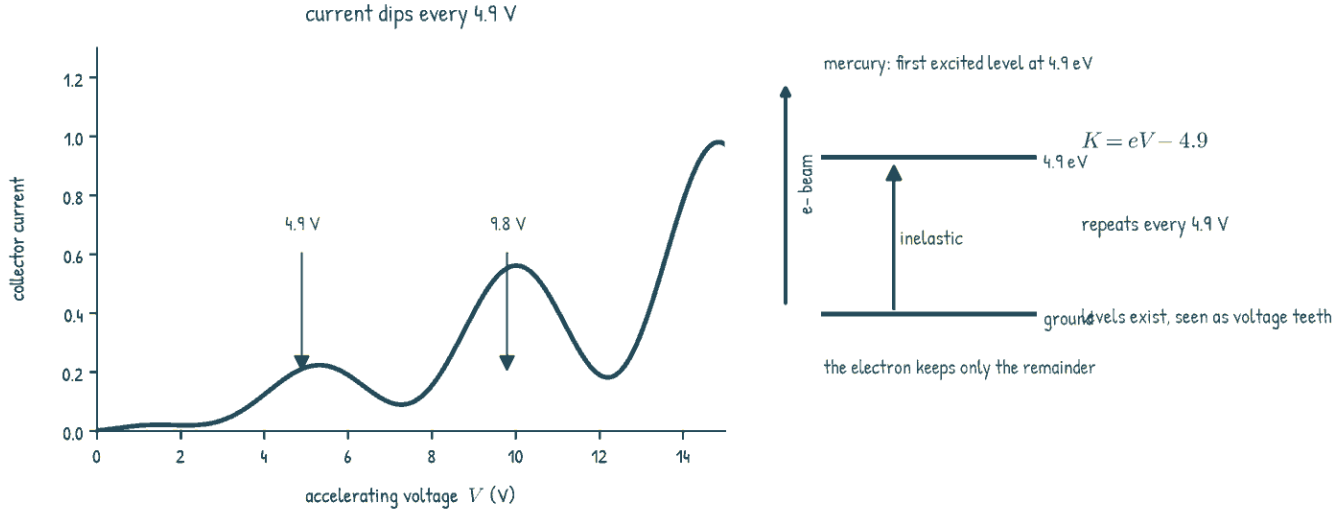
Too little vapour and the electrons never collide; too much and they scatter many times before the grid. The window is narrow, which is why the original experiment was delicate.

TRAP

reading the first dip

The first dip is shifted by the contact potential difference between the electrodes. Only the spacing is reliable.

Franck-Hertz turns a level diagram into a current-voltage curve with teeth at 4.9 V



KEEP / a dip every 4.9 V in mercury is a level diagram measured with an ammeter

How long should a wave take to eject an electron?

? Give classical physics its best case, then compare with the clock.

1

Give the wave picture every advantage

Assume the light is spread evenly, that an atom absorbs perfectly over its own cross-section, and that nothing is lost. Then the time to accumulate the escape energy is the energy needed divided by the power intercepted.

$$t = \frac{\phi}{IA} = \frac{3.2\times10^{-19}}{(10^{-3})(10^{-20})} = 3.2\times10^{4}\ \text{s}$$

→ about nine hours at a milliwatt per square metre — and a century at 10^{-8} W/m²

2

The clock says otherwise

The measured delay between switching on the light and the first electrons is below a nanosecond at every intensity tested, including beams so dim that the wave estimate runs to hours. No accumulation time exists.

$$t_{\text{observed}} \lesssim 10^{-9}\ \text{s} \quad \text{vs} \quad t_{\text{wave}} \sim 10^{4}\ \text{s}$$

→ the ratio is 10^{13}; this is not a discrepancy of factors of two

3

Why the argument is decisive rather than rhetorical

The estimate is deliberately generous. Even if the atom's cross-section were a hundred times larger the wait would be minutes, and even at sunlight intensity it would be milliseconds. Any single-photon process removes the difficulty: the energy arrives concentrated in one quantum, so there is nothing to accumulate.

→ the only escape for the wave picture is a collective absorption mechanism, which nobody has ever observed

4

A worked dim-beam estimate

Moonlight on one atom: I = 10^{-3} W/m², A = 10^{-20} m², Φ = 2 eV.

$$P = IA = 10^{-23}\ \text{W}, \qquad t = \frac{3.2\times10^{-19}}{10^{-23}} = 3.2\times10^{4}\ \text{s}$$

$$\approx 9\ \text{hours}, \quad \text{observed} < 10^{-9}\ \text{s}$$

→ thirteen orders of magnitude: the wave picture is not wrong by a factor, it is wrong

KEY RESULT

the comparison

$$t = \phi/IA$$

10^{4} s predicted, 10^{-9} s observed

one photon, one electron removes the problem

INSIGHT

why thresholds follow

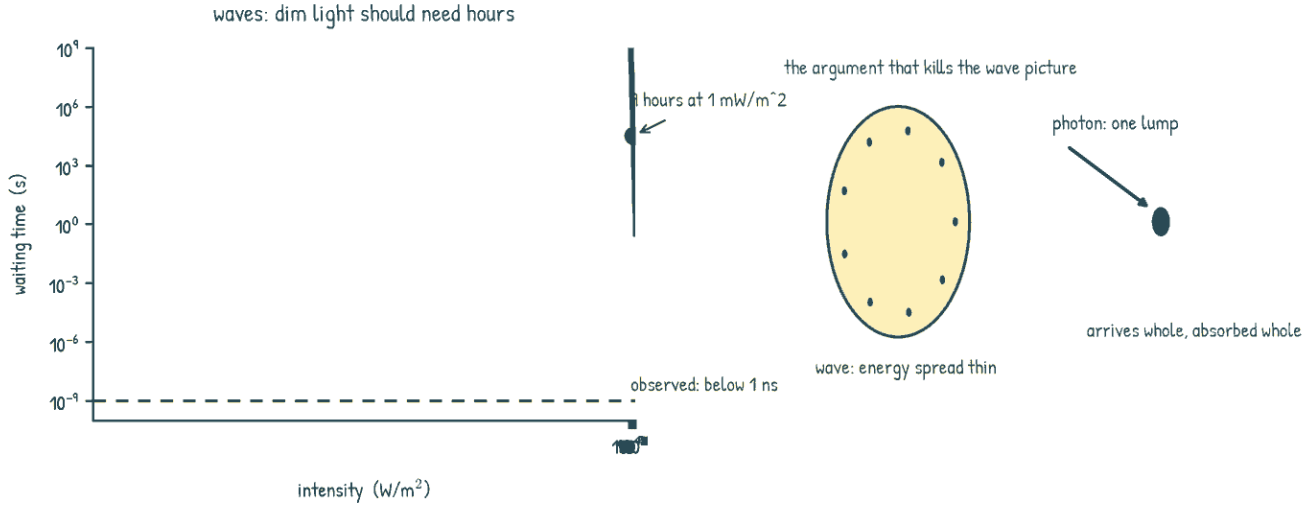
If energy arrives in quanta, then a quantum smaller than the toll simply cannot pay it — however many arrive. That is the threshold, and it needs no extra assumption.

TRAP

intensity vs energy

Saying "more intense light has more energy per photon". Intensity is photons per area per second; the energy per photon is fixed by frequency.

Classical waiting time at dim light is hours; the observed delay is nanoseconds



KEEP / hours at moonlight intensity versus nanoseconds observed: thirteen orders of magnitude

0.1 eV 511 keV 10 MeV

Which speed does a matter wave travel at?

? Two speeds, one particle, and a product that is always c².

1 Two velocities from one dispersion relation

A wave has a phase velocity ω/k and a group velocity dω/dk. For a matter wave the first is the speed of the crests and the second the speed of the envelope — the packet that actually represents the particle.

$$v_p = \frac{\omega}{k} = \frac{E}{p}, \qquad v_g = \frac{d\omega}{dk} = \frac{dE}{dp}$$

2 Substitute the relativistic dispersion relation

With $E^2 = p^2c^2 + m^2c^4$, differentiating gives $dE/dp = pc^2/E$. Dividing E by p gives the reciprocal, so the two speeds multiply to c². The group velocity comes out exactly the particle speed.

$$v_p = \frac{c^2}{v} > c, \qquad v_g = v, \qquad v_p v_g = c^2$$

→ a single infinite sine carries no information, so v_p > c violates nothing

3 Why this is not a loophole

Signal speed is the speed of a change in the envelope, which is the group velocity. For a free particle that is v < c. For light in a plasma the phase velocity also exceeds c while the group velocity is below it — the same structure, and the same reason no message outruns light.

→ a wave packet with sharp edges spreads; its front still moves at v_g

4 A worked c² product

An electron moving at 0.6c:

$$v_p = \frac{c^2}{0.6c} = 1.67c, \qquad v_g = 0.6c$$

$$v_p v_g = c^2 \ \checkmark$$

→ the crests outrun light while the particle does not; only the packet carries information

KEY RESULT
two speeds

$$v_p = E/p = c^2/v$$

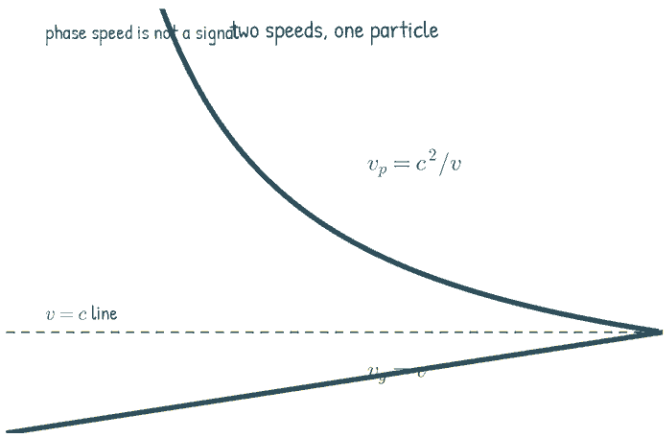
$$v_g = dE/dp = v$$

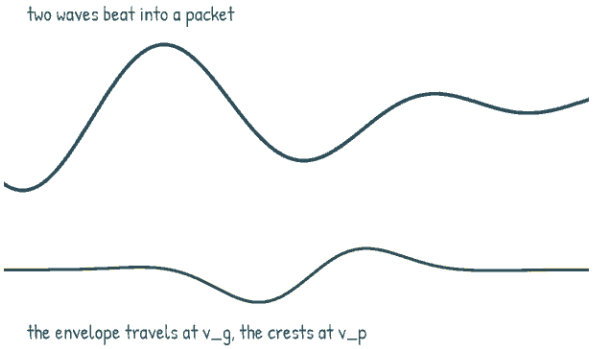
$$v_p v_g = c^2$$

GATE
when v_p > c is fine
Phase velocity is a property of a mathematical surface of constant phase, not of a physical object. No energy, no information.

TRAP
the photon limit
For m → 0 both speeds tend to c, and the distinction collapses. Do not carry the formula past that limit as if v_p were meaningful.

Matter wave: v_p v_g = c squared; only the group speed carries the particle





KEEP / phase speed c²/v exceeds c harmlessly; group speed v carries everything

0.1 eV — 1 MeV ▼ — 10 MeV

Why does a factor of two in energy swing a half-life by 10^{24}?

? Alpha decay spans thirty orders of magnitude in lifetime, yet the energies differ by a factor of two.

1 A barrier the alpha cannot climb

An alpha inside a heavy nucleus faces a Coulomb hill tens of MeV high while carrying only 4–9 MeV: classically it never leaves. It tunnels, with an exponent set by barrier width and height.

$$\log_{10}\lambda \approx a - \frac{bZ}{\sqrt{E}}$$

→ E in MeV, and the second term is of order 40 — an exponent, not a correction

2 Two nuclides show the amplifier

U-238 emits 4.27 MeV alphas and lives 4.5×10^{9} years; Po-212 emits 8.95 MeV alphas and lives 0.3 µs. The argument $1/\sqrt{E}$ changes by 30 %, and the lifetime changes by twenty-four decades.

$$\frac{1}{\sqrt{4.27}} = 0.484, \qquad \frac{1}{\sqrt{8.95}} = 0.334\ \text{MeV}^{-1/2}$$

→ a 30 % change in the argument, a 10^{24} change in the answer: exponential amplification

3 The straight line, and what it is good for

For isotopes of one element, log T against $1/\sqrt{E}$ is the Geiger–Nuttall line. Its slope grows with Z, so an unknown alpha energy lands on one element's line and not another's.

$$\log_{10} T_{1/2} = A + \frac{BZ}{\sqrt{E}}$$

→ Gamow explained in 1928 what Geiger and Nuttall had measured in 1911

4 A worked decay constant

Turn the two half-lives into rate constants, then take the ratio.

$$\lambda_{238} = \frac{\ln 2}{4.5\times10^{9}(3.156\times10^{7})} = 4.9\times10^{-18}\ \text{s}^{-1}$$

$$\lambda_{212} = \frac{\ln 2}{3\times10^{-7}} = 2.3\times10^{6}\ \text{s}^{-1}$$

→ a ratio of 4.7×10^{23} in rate for a factor 2.1 in alpha energy

KEY RESULT
the law

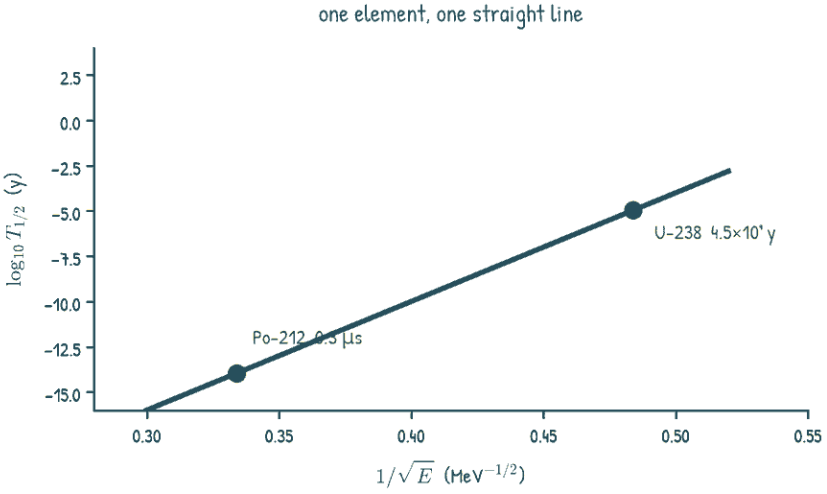
$$\log\lambda = a - bZ/\sqrt{E}$$

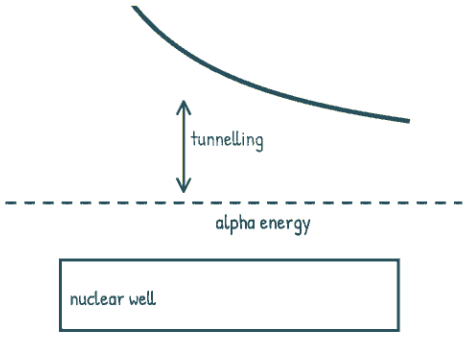
Same-Z isotopes: one straight line.
Slope ∝ Z.

GATE
what it ignores
Prefactors from nuclear structure, angular momentum hindrance, and deformation.
It is a systematics law, not a rate calculation.

TRAP
the axis
Plotting log T against E rather than $1/\sqrt{E}$ destroys the line — and the physics is in that choice of axis.

log T is linear in $1/\sqrt{E}$: a 30 % change in the argument moves the lifetime by 24 decades





KEEP / log T is linear in $1/\sqrt{E}$: the tunnelling exponent is the amplifier

Why does U-235 fission with slow neutrons and U-238 not?

? Same element, one neutron apart, completely different behaviour. The answer is a pairing bonus of 0.8 MeV.

1

Deforming the drop costs surface and saves Coulomb

Stretch the drop and its surface area grows (a cost) while the protons move apart (a gain). The barrier is the hump between: the drop must be pushed over it before Coulomb can run away.

$$B_f \approx 4\pi R^2 \gamma\left(1 - \frac{Z^2}{2A}\right)$$

→ the barrier falls as the fissility Z^2/A grows; it vanishes near $Z^2/A \approx 50$

2

Fissility is 36.0 for U-235 and 35.6 for U-238

Both sit far below the limit, so neither fissions on its own. What decides is the excitation energy supplied by the captured neutron, compared with the barrier height.

$$\frac{Z^2}{A} = \frac{92^2}{235} = 36.0, \qquad \frac{92^2}{238} = 35.6$$

→ the difference is small; the pairing term is what actually flips the outcome

3

One pairing term, two destinies

Adding a neutron to U-235 makes even-even U-236, which collects a pairing bonus of about $12/\sqrt{236} \approx 0.8$ MeV on top of the neutron's binding. U-238 plus a neutron makes odd-even U-239 with no bonus, so only a fast neutron's kinetic energy closes the gap.

$$E^*_{236} \approx 6.5\ \text{MeV} > B_f, \qquad E^*_{239} \approx 4.8\ \text{MeV} < B_f$$

→ thermal neutrons for U-235, fast neutrons for U-238 — one line of liquid-drop arithmetic

4

A worked pairing bonus

The excitation of U-236 against its barrier.

$$\delta = \frac{12}{\sqrt{236}} = 0.78\ \text{MeV}$$

$$E^*_{236} = 5.7 + 0.78 = 6.5\ \text{MeV} > B_f \approx 6\ \text{MeV}$$

→ U-239 gets 4.8 MeV and stays below the barrier: no pairing bonus, no thermal fission

KEY RESULT

the numbers

Barrier ≈ 6 MeV for both.

Excitation 6.5 vs 4.8 MeV.

Fissility 36.0 vs 35.6.

INSIGHT

why even-even pays

Nucleons pair with opposite spins; an unpaired nucleon costs energy. Even-even nuclei are the tightly bound ones.

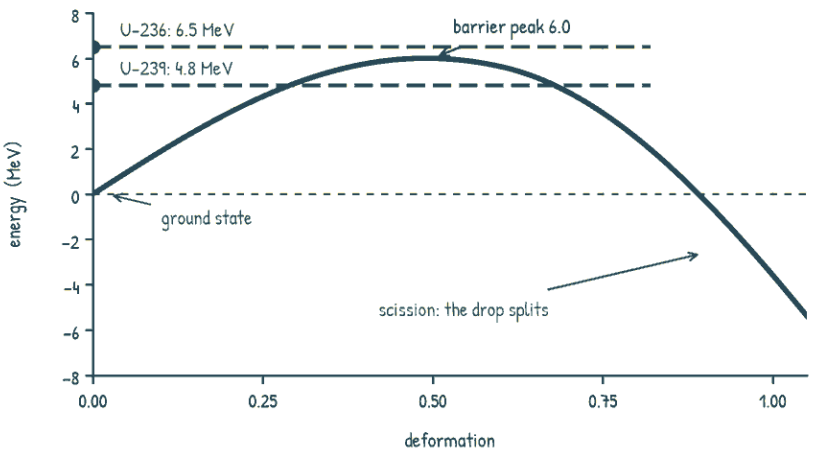
TRAP

the fissility shortcut

Using Z^2/A alone to predict which isotope fissions. It sets the barrier's trend, not the outcome for one neutron.

Deformation costs surface and saves Coulomb; pairing decides which nucleus gets over the hump

one barrier; the excitation decides who clears it



surface cost, then Coulomb gain



KEEP / the barrier is ~6 MeV for both; pairing decides who clears it

Can you derive the valley rather than remember it?

? Maximise the liquid-drop binding over charge at fixed mass. One derivative gives the stability line.

1 **Only two terms care about Z**

At fixed A the volume and surface terms are constants, and pairing is a small step. The competition is between the Coulomb term (grows as Z^2) and the asymmetry term (grows as $(A-2Z)^2$).

$$\frac{\partial B}{\partial Z} = -2a_c\frac{Z}{A^{1/3}} - 2a_a\frac{2(A-2Z)}{A} = 0$$

→ the asymmetry term is written so that N − Z = A − 2Z

2 **Solve it**

Rearranging gives the floor explicitly. Small A returns Z = A/2; large A pushes the floor toward neutron-rich nuclei because the Coulomb tax grows faster than the asymmetry tax.

$$Z_0 = \frac{4a_a A}{8a_a + a_c A^{2/3}} = \frac{A}{2 + \frac{a_c}{2a_a}A^{2/3}}$$

→ a_c/2a_a = 0.71/46.4 = 0.0153, the 0.0154 of the formula sheets

3 **Test it where the data are**

At A = 100 the formula gives 42.9; the stable isobars are Mo and Ru, Z = 42–44. At A = 238 it gives 92.0 and uranium is at 92. The formula is a mean trend: shell closures dent it, which is why the real chart has kinks.

$$Z_0(100) = 42.9, \qquad Z_0(238) = 92.0$$

→ the liquid drop is the climate; the shells are the weather

4 **A worked coefficient**

Check the 0.0154 in the valley formula.

$$\frac{a_c}{2a_a} = \frac{0.71}{2(23.2)} = 0.0153$$

$$Z_0(56) = \frac{56}{2 + 0.0153(56)^{2/3}} = 27.0$$

→ iron sits at Z = 26: the liquid drop is one unit high, and the shell closure explains it

KEY RESULT

the derivative

$$Z_0 = \frac{A}{2 + 0.0154A^{2/3}}$$

N = Z at small A.

N/Z → 1.5 at A = 238.

GATE

what is dropped

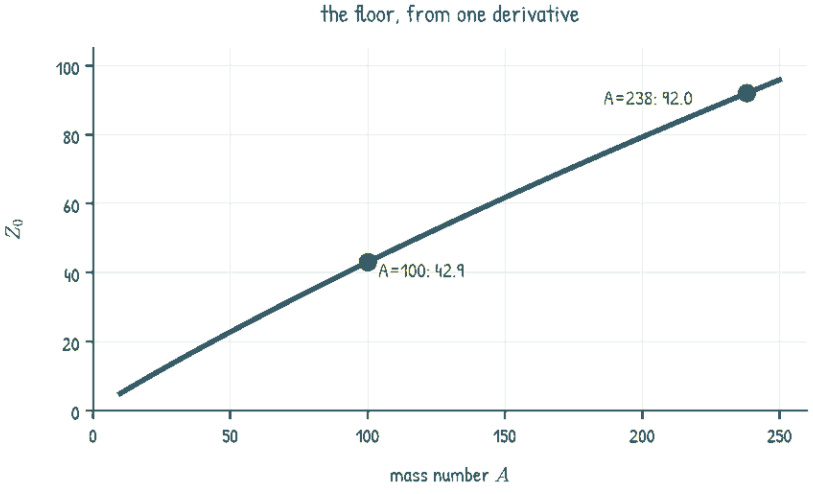
Pairing's derivative is ignored and shell effects are absent, so the formula misses magic-number nuclei by several units of Z.

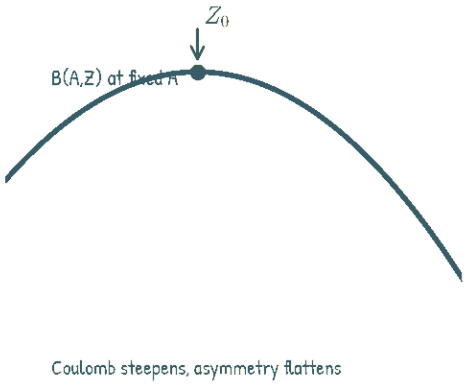
TRAP

differentiating A

Treating A as a variable. The valley is defined at fixed A — that is the whole point of an isobar.

Differentiate B over Z at fixed A: the valley floor follows, and shells dent it





KEEP / differentiate B(A,Z): Z₀ = A/(2 + (a_c/2a_a)A^{2/3})

0.1 eV 1 MeV 10 MeV

How long does a daughter take to peak?

? Solve the two-step chain properly, then read the equilibrium and the peak time off the solution.

1 Solve the pair with an empty daughter

Substitute the parent's exponential into the daughter's equation and integrate with $N_2(0) = 0$. The result is a difference of two exponentials: the daughter's own decay minus the parent's supply.

$$N_2(t) = \frac{\lambda_1 N_1(0)}{\lambda_2 - \lambda_1}\left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right)$$

→ when $\lambda_2 > \lambda_1$ the second exponential dies first and the parent survives

2 Late times leave one exponential

After the daughter's own transient has died away, the bracket approaches e^{-λ₁t} and the ratio of activities becomes a constant. That constant is the equilibrium, and it is 1 in the secular limit.

$$\frac{A_2}{A_1} \to \frac{\lambda_2}{\lambda_2 - \lambda_1} \to 1 \quad \text{as } \lambda_1 \ll \lambda_2$$

→ transient ratio 1.5 when $\lambda_2 = 3\lambda_1$; secular ratio 1 when the parent is very long-lived

3 Differentiate again for the peak

Setting the derivative of N_2 to zero gives the time of maximum daughter activity. It is the crossing of the two exponentials — the moment the daughter's production and decay balance at their largest.

$$t_{\max} = \frac{\ln(\lambda_2/\lambda_1)}{\lambda_2 - \lambda_1}$$

→ the peak matters for a generator: elute the daughter at the maximum, not later

4 A worked maximum

$\lambda_1 = 0.02\ \text{h}^{-1}$, $\lambda_2 = 0.10\ \text{h}^{-1}$: ratio and peak time.

$$\frac{A_2}{A_1} = \frac{0.10}{0.08} = 1.25$$

$$t_{\max} = \frac{\ln 5}{0.08} = 20.1\ \text{h}$$

→ the daughter peaks at 20 h, then falls with the parent's 34.7 h half-life

KEY RESULT

the three results

$$N_2(t) \propto e^{-\lambda_1 t} - e^{-\lambda_2 t}$$

$$A_2/A_1 = \lambda_2/(\lambda_2 - \lambda_1)$$

$$t_{\max} = \ln(\lambda_2/\lambda_1)/(\lambda_2 - \lambda_1)$$

GATE

limits behave

$t \to 0$: $N_2 \to 0$ with zero slope.

$\lambda_1 \to 0$: ratio → 1, secular.

Dimensions: t_max is a time.

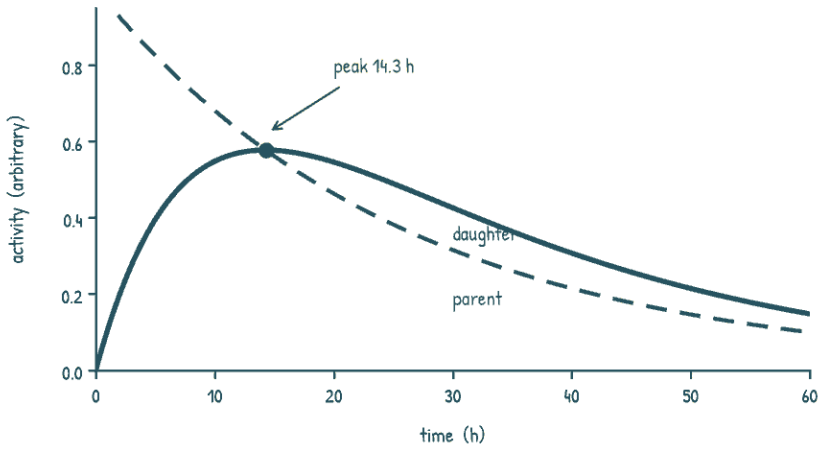
TRAP

which exponential

Subtracting the parent from the daughter (sign flip) gives a negative activity. The daughter term always decays faster.

A_2/A_1 settles at $\lambda_2/(\lambda_2 - \lambda_1)$; the peak is where production and decay balance

rise, peak, then follow the parent



equilibrium ratio

transient: 1.5

secular: 1.0

the longer-lived parent sets the tempo

KEEP / A_2 peaks at $\ln(\lambda_2/\lambda_1)/(\lambda_2 - \lambda_1)$, then follows the parent

How many solar neutrinos cross your thumbnail each second?

? The Sun's luminosity fixes its fusion rate, and the fusion rate fixes the neutrino flux at Earth.

KEY RESULT
the chain of division
L → helium rate → neutrino rate → flux at 1 AU.
26.7 MeV and two neutrinos per helium.

GATE
the assumptions
Steady Sun, p-p chain dominant (CNO is a few per cent), and no neutrino loss inside the Sun.

TRAP
one neutrino or two
Counting one neutrino per helium halves the flux; the chain emits two, one per positron annihilation step.

1 Luminosity counts helium nuclei

Every 26.7 MeV released corresponds to one helium nucleus made and two neutrinos emitted. Dividing the luminosity by that energy gives the helium rate directly, with no model of the core needed.

$$\dot{N}_{\mathrm{He}} = \frac{3.83\times10^{26}}{26.7\times1.602\times10^{-13}} = 8.9\times10^{37}\ \mathrm{s}^{-1}$$

→ twice that many neutrinos: the chain emits two per helium

2 Spread them over a sphere

At Earth the flux is that rate divided by 4π times the astronomical unit squared. The number is enormous, and it is the number every solar-neutrino experiment had to explain.

$$\phi_\nu = \frac{1.8\times10^{38}}{4\pi(1.496\times10^{11})^2} = 6.4\times10^{14}\ \mathrm{m}^{-2}\mathrm{s}^{-1}$$

→ through one square centimetre: 6×10¹⁰ per second, through a thumbnail every second

3 The detected flux is a third of this

The measured flux on Earth is about a third of the prediction, and the resolution is neutrino oscillation: electron neutrinos convert into muon and tau flavours on the way out. The total flux is right; only the flavour mix changes.

$$\phi_{\nu_e} \approx \tfrac{1}{3}\phi_\nu \quad \text{(oscillated)}, \qquad \phi_{\text{total}}\ \text{conserved}$$

→ a flux measurement is a flavour measurement, not a luminosity test

4 A worked flux

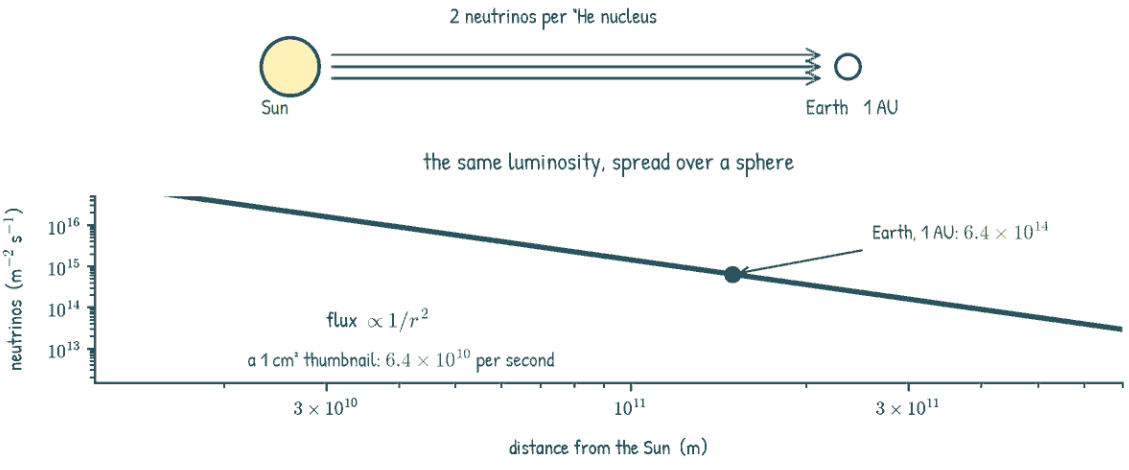
Helium rate, neutrino rate, and the flux at 1 AU.

$$\dot{N}_{\mathrm{He}} = \frac{3.83\times10^{26}}{26.7(1.602\times10^{-13})} = 8.9\times10^{37}\ \mathrm{s}^{-1}$$

$$\phi = \frac{2(8.9\times10^{37})}{4\pi(1.496\times10^{11})^2} = 6.4\times10^{14}\ \mathrm{m}^{-2}\mathrm{s}^{-1}$$

→ 6×10¹⁰ per second through a square centimetre, day and night

Luminosity counts helium nuclei; helium counts neutrinos; geometry turns that into a flux



KEEP / L/26.7 MeV gives 1.8×10³⁸ neutrinos per second; 6×10¹⁴ m⁻²s⁻¹ at Earth

How much fuel does a reactor eat in a day?

? A gigawatt is a mass rate in disguise, and ten half-lives is an engineering unit of forgetfulness.

1

Energy per day, then mass per day

A gigawatt thermal for a day is 8.6×10¹³ J. Dividing by the energy released per gram of U-235 gives about a kilogram — the whole fuel bill of a large reactor fits in a briefcase.

$$E_{\text{day}} = 10^9(86400) = 8.6\times10^{13}\ \text{J}, \qquad m = \frac{8.6\times10^{13}}{8.2\times10^{10}} = 1.05\ \text{kg}$$

→ at 33 % efficiency an electrical gigawatt needs three times that, about 3.2 kg/day

2

Why a kilogram is not a gram

Only the U-235 fissions; natural uranium is 0.7 % U-235, and a reactor enriches it to a few per cent. The mass of uranium consumed is therefore tens of times the mass fissioned, and most of the spent fuel is still uranium.

$$m_{\text{U}} \approx \frac{1.05}{0.04} \approx 26\ \text{kg/day}\quad \text{fed}$$

→ at 4 % enrichment; the difference between burned and fed is the fuel cycle

3

Ten half-lives is the storage shorthand

Each half-life removes a factor two; ten of them remove a factor 1024. That is why "store it for ten half-lives" is the phrase in waste policy: it turns an alarming activity into a small fraction, without ever reaching zero.

$$2^{-10} = 9.8\times10^{-4} \approx 0.1\,\%$$

→ a 30-year isotope stored 300 years is ten half-lives — 0.1 % of the original activity

4

A worked fuel bill

1 GW thermal for one day, and the activity after ten half-lives.

$$m = \frac{10^9(86400)}{8.2\times10^{10}} = 1.05\ \text{kg}$$

$$2^{-10} = 9.8\times10^{-4}$$

→ a kilogram a day, and a thousandfold drop per ten half-lives — both are rules of thumb

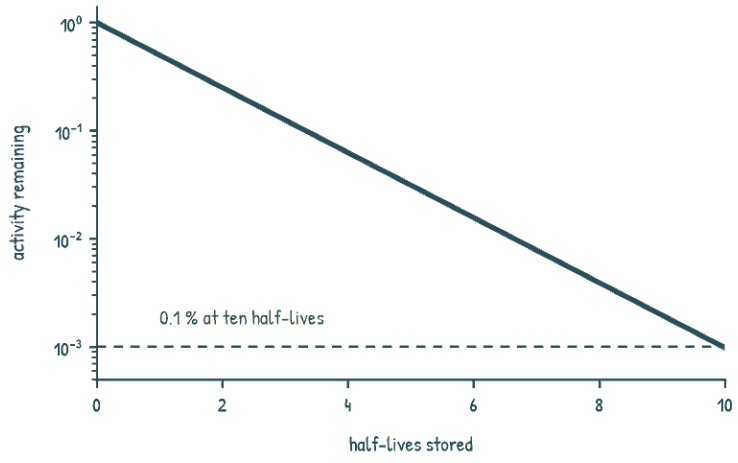
KEY RESULT
the numbers
1.05 kg fissioned per GW·day (thermal).
≈26 kg of 4 % fuel fed per day.
$2^{-10} \approx 0.1\,\%$.

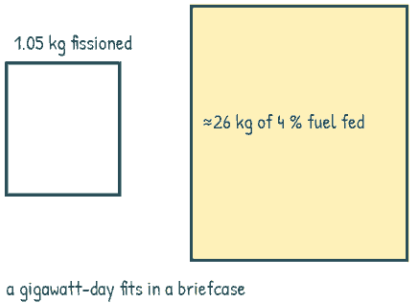
GATE
what this ignores
Burn-up, conversion ratio, decay heat, and the activity of the fission products — second-order for fuel mass, first-order for safety.

TRAP
thermal versus electric
Quoting 1.05 kg/day for a 1 GW *electric* plant. At 33 % efficiency it is three times that.

1.05 kg fissioned per GW·day; the fuel fed is tens of times more because the enrichment is small

the storage rule of thumb





a gigawatt-day fits in a briefcase

KEEP / 1.05 kg per day per GW thermal; ten half-lives leaves 0.1 %

How long must you count to know the rate to 1 %?

? Radioactive counts arrive like coin flips. The square-root tax sets every measurement time in the lab.

1 Poisson statistics in one line

Counts in a fixed window are independent events with a fixed mean, so the variance equals the mean. The absolute uncertainty is √N and the fractional uncertainty is 1/√N — the only error bar that gets relatively smaller as you count more.

$$\sigma_N = \sqrt{N}, \qquad \frac{\sigma_N}{N} = \frac{1}{\sqrt{N}}$$

→ 10 000 counts carry 1 %; 100 counts carry 10 %

2 Plan the time backwards from the precision

At 100 counts per minute, 1 % needs 10 000 counts, which is 100 minutes. A weak sample at 1 count per minute needs 10 000 minutes — a week of counting for the same sentence.

$$N = \left(\frac{1}{\epsilon}\right)^2 = 10^4 \text{ for } \epsilon = 1\,\%, \qquad t = \frac{N}{r}$$

→ to halve the error you need four times the counts, hence four times the time

3 Dead time and background bend the rule

A Geiger counter is blind for a short dead time after each pulse, so at high rates it undercounts; the correction is n/(1 − nτ_d). Background counts add in quadrature, so a weak source must be counted with and without.

$$n_{\text{true}} = \frac{n_{\text{obs}}}{1 - n_{\text{obs}}\tau_d}, \qquad \sigma_{\text{net}} = \sqrt{N_s + 2N_b}$$

→ at 1 % dead time and 10^4 counts per second the correction is already 1 %

4 A worked counting plan

A source gives 100 counts per minute. Time for 1 %?

$$N = \frac{1}{(0.01)^2} = 10^4 \;\Rightarrow\; t = \frac{10^4}{100} = 100 \text{ min}$$

$$\text{for } 0.5\%: \; N = 4 \times 10^4, \; t = 400 \text{ min}$$

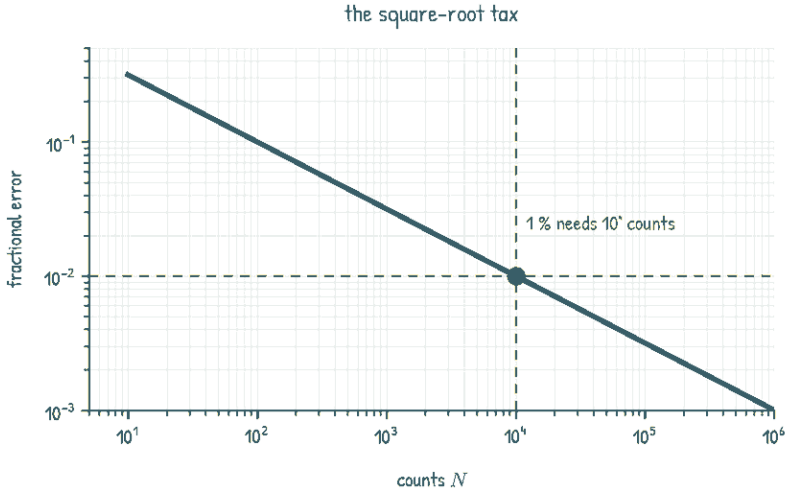
→ four times the time for half the error; the tax is unavoidable

KEY RESULT
the tax
$\sigma/N = 1/\sqrt{N}$
1 % → 10^4 counts.
4× time halves the error.

GATE
when Poisson holds
Independent decays, a stable rate, and a window short compared with the half-life.

TRAP
the doubled background
Subtracting background without adding its uncertainty — the net error is √(N_s + 2N_b), not √N_s.

σ = √N; to halve the fractional error you need four times the counts and four times the time



rate 100 min⁻¹

1 % → 100 min

0.5 % → 400 min

0.1 % → 10^4 min

four times the time halves the error

KEEP / σ = √N, so 1 % needs 10^4 counts and four times the time halves the error

0.1 eV 1 MeV 10 MeV

How do you date a rock without knowing what it started with?

? Plot one isotope ratio against another. Every mineral of the same rock falls on one line, and the slope is the age.

1 Write the daughter as initial plus produced

The daughter isotope present now is whatever was there at birth plus what the parent has since produced. Dividing by a stable reference isotope turns abundances into ratios that can be measured precisely.

$$\frac{D}{S} = \left(\frac{D}{S}\right)_0 + \frac{P}{S}(e^{\lambda t} - 1)$$

→ D/S and P/S are measured; the unknowns are the intercept and the slope

2 The line removes the unknown

Different minerals in one rock have different P/S but the same age and the same initial D/S. Plotting D/S against P/S gives a straight line for every mineral: its slope is e^{λt} − 1 and its intercept is the initial ratio.

$$\text{slope} = e^{\lambda t} - 1 \;\Rightarrow\; t = \frac{\ln(1 + \text{slope})}{\lambda}$$

→ scatter off the line is the diagnostic: it says the rock was disturbed

3 Read an age off a slope

For Rb-Sr, λ = 1.42×10⁻¹¹ per year. A slope of 0.05 gives ln(1.05)/λ = 3.4×10⁹ years, which is the age of an Archean rock.

$$t = \frac{\ln 1.05}{1.42 \times 10^{-11}} = 3.4 \times 10^{9}\ \text{y}$$

→ the same arithmetic as carbon dating, with a different constant and no initial-ratio guess

4 Two minerals, one slope

Rb-Sr, λ = 1.42×10⁻¹¹ y⁻¹. Mineral A: 2.0 and 0.720. Mineral B: 5.0 and 0.750.

$$\text{slope} = \frac{0.750 - 0.720}{5.0 - 2.0} = 0.0100$$

$$t = \frac{\ln 1.010}{1.42 \times 10^{-11}} = 7.0 \times 10^{8}\ \text{y}$$

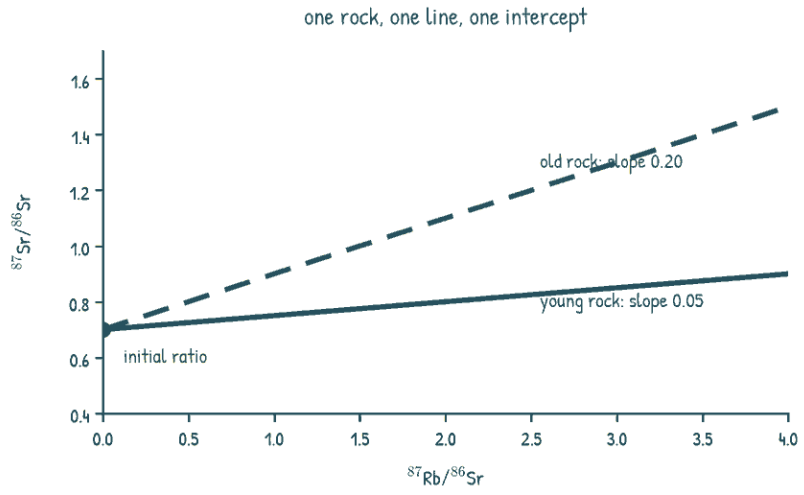
→ check the intercept: 0.720 − 0.0100(2.0) = 0.700, the same for B — one age, one initial ratio

KEY RESULT
the line
D/S against P/S.
Slope = e^{λt} − 1.
Intercept = initial ratio.

GATE
the isochron assumptions
One age for all minerals, one initial ratio, a closed system since formation.

TRAP
the model age
Using a single mineral and an assumed initial ratio gives a model age, not an isochron age. It is only as good as the assumption.

Plot daughter/reference against parent/reference: slope gives the age, intercept the start



$\text{slope} = e^{\lambda t} - 1$

slope 0.05 → 3.4 Gyr

intercept = initial ratio

scatter off the line = disturbed rock

KEEP / slope = e^{λt} − 1; the intercept absorbs the unknown initial daughter

0.1 eV — 1 MeV ▼ — 10 MeV

Why do slow neutrons diffract from crystals?

? The same wave logic as the electron: compare the wavelength with the spacing. The spacing that matters is the lattice's.

1 A thermal neutron has an angstrom wavelength

A 25 meV neutron has λ = h/√(2mE) = 1.8 Å — the scale of atomic planes. That is why neutron diffraction needs no accelerator.

$$\lambda = \frac{h}{\sqrt{2m_n E}} = 1.8\ \text{Å} \quad \text{at } 25\ \text{meV}$$

→ the same formula as the electron's 1.226/√V nm, with the neutron mass and energy

2 Two rulers, one formula

Inside a nucleus the nucleon spacing is about 1.8 fm, a thousand times smaller. A thermal neutron with a 1.8 Å wave cannot resolve that structure — it sees the whole nucleus as a point scatterer. The physics did not change; the ratio λ/d did.

$$d_{\text{nucleon}} \approx \left(\frac{1}{0.17\ \text{fm}^{-3}}\right)^{1/3} \approx 1.8\ \text{fm} \ll \lambda$$

→ high-energy electrons (pm waves) are needed to see inside a nucleus, for exactly this reason

3 Why neutrons rather than X-rays

X-rays scatter from electrons, so hydrogen is nearly invisible. Neutrons scatter from nuclei, so light atoms show and isotopes contrast — same Bragg law, a different probe.

$$2d\sin\theta = n\lambda \quad (\text{Bragg}), \qquad \lambda = 1.8\ \text{Å} \Rightarrow \theta \text{ of order } 10^\circ$$

→ magnetic moments also scatter neutrons, which is why they map spin structures

4 A worked wavelength

A 25 meV neutron, and the energy needed to reach 1 fm.

$$\lambda = \frac{h}{\sqrt{2m_n E}} = 1.81\ \text{Å} \quad \text{at } 25\ \text{meV}$$

$$E_k = \sqrt{(pc)^2 + (m_n c^2)^2} - m_n c^2 = 616\ \text{MeV} \quad \text{for } \lambda = 1\ \text{fm}$$

→ at that momentum the non-relativistic $h^2/2m\lambda^2$ would say 818 MeV — the relativistic form is the one to use

KEY RESULT
the comparison
λ(25 meV) = 1.8 Å.
Lattice spacing ≈ 1.8 Å → diffraction.
Nucleon spacing 1.8 fm → point scatterer.

GATE
when diffraction works
λ comparable to the spacing. Much shorter gives a point image; much longer gives no Bragg condition at all.

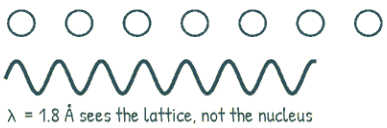
TRAP
mixing the rulers
Using a thermal neutron to probe nuclear structure, or an MeV neutron to probe a crystal — the wavelength is wrong in both directions.

Match the wavelength to the spacing: thermal neutrons see crystals, high-energy probes see nuclei

nucleus: spacing 1.8 fm



crystal: spacing 1.8 Å



λ = 1.8 Å sees the lattice, not the nucleus

25 meV → 1.8 Å

616 MeV → 1 fm

non-relativistic: 818 MeV (wrong)

2.5×10^{10} in energy

same λ = h/p, different probe

KEEP / 1.8 Å neutrons match atomic spacings; inside a nucleus the same wave sees a point

0.1 eV — 1 MeV ▼ — 10 MeV

How long can the Sun keep shining?

? Ten per cent of its hydrogen, 0.7 per cent of the mass of that fuel, and a luminosity you can measure.

1 Count the fuel, then the energy per gram

Only the core hydrogen burns, and only 0.7 % of the fused mass becomes energy. Multiplying the available mass by 0.007c² gives the total energy budget; dividing by the luminosity gives the lifetime.

$$E \approx 0.1(0.7M_\odot)(0.007c^2) = 8.7 \times 10^{43}\ \text{J}$$

→ the numbers: 10 % of the hydrogen, 70 % hydrogen by mass, 0.7 % mass-to-energy

2 Divide by the luminosity

The Sun radiates 3.83×10²⁶ W, so the budget lasts about 7×10⁹ years on the main sequence — and the independent mass-loss rate L/c² times that lifetime returns the burned mass, a useful cross-check.

$$t = \frac{8.7 \times 10^{43}}{3.83 \times 10^{26}} = 2.3 \times 10^{17}\ \text{s} \approx 7 \times 10^{9}\ \text{y}$$

→ the Sun is middle-aged; the first weak step in the p-p chain, not the fuel, sets the pace

3 The same liquid drop says where the table ends

The fission barrier vanishes when the Coulomb gain from deformation beats twice the surface cost, near Z²/A ≈ 50. With A ≈ 2.5Z that is Z ≈ 125; shell corrections prop up superheavies a little beyond, but not indefinitely.

$$\frac{Z^2}{A} \approx 50, \quad A \approx 2.5Z \ \Rightarrow\ Z \approx 125$$

→ bismuth (83) is the last fully stable element; beyond it, alpha decay is the leak

4 A worked solar budget

Ten per cent of the Sun's hydrogen, 0.7 % mass-to-energy.

$$E = 0.1(0.7)(1.989 \times 10^{30})(0.007)(9 \times 10^{16}) = 8.8 \times 10^{43}\ \text{J}$$

$$t = \frac{8.8 \times 10^{43}}{3.83 \times 10^{26}} = 2.3 \times 10^{17}\ \text{s} = 7.3 \times 10^{9}\ \text{y}$$

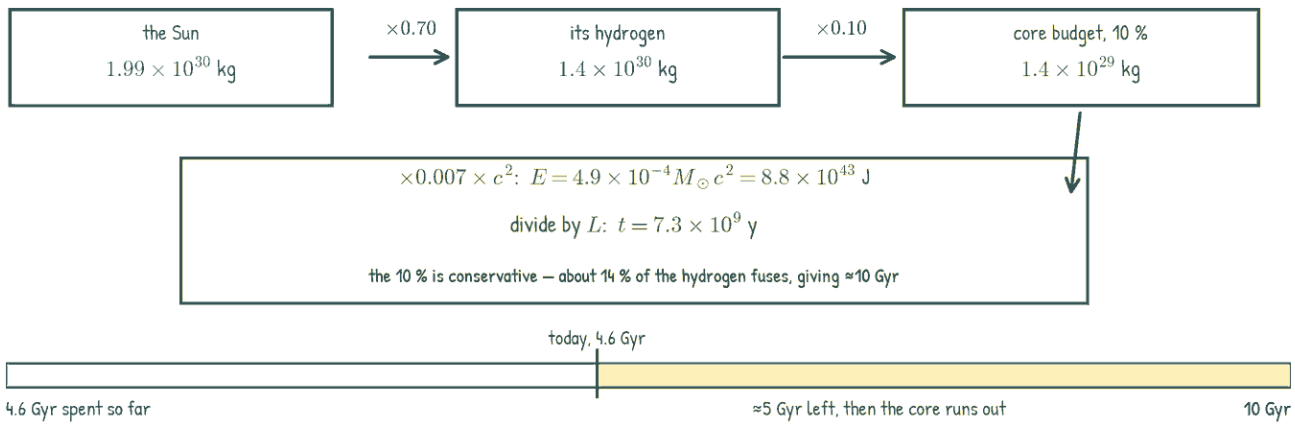
→ the Sun has burned about half its hydrogen; it is middle-aged, not ancient

KEY RESULT
the budget
10 % of the hydrogen is available.
0.007 of the mass becomes energy.
L = 3.83×10²⁶ W → ≈7×10⁹ y.

GATE
what is left out
Mass loss, core mixing, the red-giant phase, and the fact that the last of the hydrogen burns inefficiently.

TRAP
all the hydrogen
Using the whole mass, or the whole hydrogen, in the budget overestimates the lifetime by a factor of ten.

0.1 × 0.7 × 0.007 × M☉c² divided by L gives about 7 billion years



KEEP / 0.07 × 0.1 M☉ × 0.007c² / L ≈ 10 Gyr; and the drop says where the table ends